CAPITAL STRUCTURE

14

Whilst debt offers distinct tax advantages, the global financial crisis brought to the fore the risks of high levels of debt. Banks in the USA and UK were highly geared and many could not survive the perfect financial storm that hit capital markets in 2008. Comparatively, corporate firms generally had low debt–equity ratios and most were able to survive the turmoil arising from the crisis. Low debt enabled them to operate and sometimes borrow to survive a difficult operating environment. Highly geared individual mortgage holders were not so lucky. Therefore, investing in highly geared firms is risky. Low debt and access to cash offers firms the flexibility to survive financial storms and also enables firms to invest quickly if opportunities arise.

South African firms generally have very low debt–equity ratios although increasingly firms are engaging in share buybacks to change their capital structures. Just over 25 years ago, interest rates were raised to over 24% and in the crisis in 2008 even good businesses could not acquire financing from the banks. Management does not forget. As Mark Twain wrote: a cat that has sat on a hot stove will not sit on a hot stove again, but neither will it sit on a cold stove. This cat will simply not like stoves. In 2024, interest rates remain high relative to other developed economies. This reflects perhaps the reluctance of South African companies to make more use of debt, as management have lived with high and volatile interest rates in prior years. Warren Buffett has stated that "a base business can not be transformed into a golden business by tricks of accounting or capital structure". He prefers to invest in companies with low borrowings. Yet many companies are of the view that an appropriate target capital structure will reduce their cost of capital. For example, Distell in its 2022 annual financial statements asserts that an objective of management is to "maintain an optimal capital structure to reduce the cost of capital". Sasol reports that its target debt to equity ratio is 30% and Net Debt: EBITDA should be less than 1.5.

Capital structure theory is based on the research of Modigliani and Miller (MM) who made two important propositions; firstly, the value of a company is determined by the earning power and the riskiness of its assets, not by how the assets are financed

and secondly, that the cost of equity increases with financial leverage so that you cannot reduce your cost of capital by taking on more debt. Miller used to explain MM's first proposition by linking it to the story attributed to the famous baseball player, Yogi Berra, who requested the pizza man to cut up his pizza into eight pieces rather than four because he was feeling hungry.

MM made certain assumptions, principally about there being no tax and so this may change our conclusions. Then there are agency issues in that the interests of shareholders and bondholders may diverge particularly for highly levered firms. In this chapter, we will explore all these issues.

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Explain how leverage increases returns to shareholders.
- Explain how debt increases financial risk.
- Show why there is no optimal capital structure under certain assumptions.
- Show how the inclusion of tax, financial distress and other costs alters the optimal capital structure.
- Apply the theory of capital structure to practical situations and explain the trade-off theory.
- Explain the pecking order approach.
- Show the effect of inflation on the cost of debt.
- Explain the relationship between an optimal capital structure and the need for flexibility.
- Show how the tax shield that arises through the use of debt increases the attractiveness of utilising debt.
- Indicate how the signalling theory can explain capital structure decisions.
- Understand how debt may be employed to resolve agency issues between management and shareholders.

- Determine the conflicts that can arise between shareholders and bondholders for highly levered firms and understand the role that loan covenants play in resolving these agency issues.
- Determine the value of tax shields arising from changes in a firm's capital structure and determine the impact on a firm's cost of capital.
- Evaluate the possible effects that personal taxes can have on the capital structure decision.
- Explain how behavioural factors can influence a firm's capital structure decisions.

INTRODUCTION

One of the areas in financial management that has attracted a great deal of academic research is the relationship between a firm's cost of capital, its value, and its capital structure. As was discussed in Chapter 7, the lower the firm's cost of capital, the higher the value of the firm. However, it is much less clear whether differing capital structures can lead to differing costs of capital. If this relationship holds, financial managers should determine the capital structure which leads to the lowest cost of capital and which consequently maximises the value of the firm.

Consider the framework presented in the first chapter in which financial management was seen as the catalyst between investment opportunities on the one hand and sources of finance on the other. This is reflected in the balance sheet which shows assets on the one hand, and the financing of those assets on the other. A business may be viewed as a portfolio of all its assets. The financing of these assets can be considered as a separate issue.

The way in which the financing is arranged is a strategic financial decision, the result of which is the capital structure. The management must decide on the level of debt, relative to equity, that it wants to maintain, and it can decide on the types of debt and the types of equity that it plans to hold. These decisions are influenced by the financial policy that the management chooses and their ability to source the desired type of financing they choose. The policy, in turn, is influenced by the risk profile that management adopts in order to maximise the value of the firm.

1 RISK PROFILE

Financial managers should strive to maintain a balance in the risks to which the company is exposed. There is a convention that an optimal strategy is to match long-term assets with long-term finance, and short-term assets with short-term finance. This is known as maturity matching. While this may often be a very sensible way to arrange finance, the company may choose to arrange it differently. This decision is a consequence of the policy that the company adopts towards risk. It also depends on other risks to which it is exposed. The starting point in evaluating the risk profile of a company is the evaluation of business risk and financial risk.

Business risk

Business risk is *risk relating to the operating activities of the company*. This is often reflected by the company's level of operating leverage which was discussed in Chapter 3. Business risk relates to the nature of the industry and the company's operations. For example, an organisation such as a retail grocery chain has a low business risk because, irrespective of the economic climate, it deals in basic necessities. Its products are the last to be affected in recessionary times. People still need to attend to the basic necessities of life, such as nutrition and healthcare. Likewise, in boom periods the effect on basic products is not as great as it would be on luxury goods. A grocery chain may still have to face competitive threats from other companies in the sector. At the other end of the spectrum is the building and construction industry, which suffers severely during periods of economic recession, but is very favourably placed to take advantage of economic upswings. This is one of the Page 14-3 reasons that investors are encouraged to diversify their investments, as is discussed in detail in Chapter 4. In times of uncertainty or recession investors fall back into so called defensive stocks that have a low business risk due to the continuing demand for their products, no matter what the state of the economy. In better times investors seek out the stocks that will benefit from increased discretionary spending and thus have a higher business risk. This is the classical risk and return nexus, which is one of the cornerstones of finance.

Financial risk

Financial risk is primarily influenced by the *balance between debt and equity*, i.e. by its capital structure. It consists of two parts: *interest risk and capital risk*. The interest risk is the risk of not being able to cover the fixed interest charge. The capital risk is

that the firm will not be able to pay back its borrowings on time and may then be placed into bankruptcy or business rescue.

Equity

The equity of a company is the *funds supplied by the shareholders*. The reward for these funds is the dividend paid to the shareholders and the capital growth in the value of the shares. The risk to the company is low because, in the event of economic hardship, the dividend need not be paid. On the other hand, when economic times are good, the shareholders benefit greatly as the dividend is higher, or the capital growth is higher.

Debt

Debt includes all borrowed funds. These funds usually have contractual interest commitments and capital repayments to be considered. Debt can be used to the advantage of the company through the effect of gearing. By using debt, the organisation exposes itself to financial risk. In the event of economic decline, interest payments and capital repayments must still be made, and if the company has not generated sufficient income it may be unable to meet these payments. Given that the exposure to debt increases risk, the extent of the risk can be managed by the choice of the form in which the debt is placed. As there is increased risk to the company the shareholders must be rewarded for this higher risk, but more of this later.

- *Long term or short term.* Short-term debt is usually riskier because the commitments have to be met rapidly, whereas in the case of long-term debt there is a longer period in which to plan for repayment. Short-term debt has the advantage of allowing the borrower greater flexibility to react to changing circumstances such as falling interest rates.
- *Secured or unsecured.* Whether the company's debt is secured or unsecured is influenced by the lender's attitude as the loan may be granted only on condition that it is secured.

It is also a matter of policy as to whether the company is prepared to subject itself to the risk of its assets being provided as security. Secured debt is associated with higher risk because there is less flexibility on the part of the borrowing company.

A company which is not subject to huge fluctuations in its earnings pattern may be prepared to be exposed to greater financial risk. On the other hand, a company

subject to volatile trading conditions would be unwise to take a position of high financial risk because, in the event of an economic downturn, it could lead to financial distress. Technology and biotech start-up firms have often only used equity financing. The companies did not have assured cash flows nor established business models. Their ability to pay was constrained. Those that did not establish a viable cash flow found the debt and equity markets hostile to further funding and disappeared.

2 LEVERAGE (GEARING)

Using relatively more debt in the capital structure is known as leverage (or gearing). Leverage is intended to increase the return on shareholders' funds in exchange for greater financial risk. Page 14-4

Impact on earnings

The use of financial leverage - i.e. the use of debt rather than equity for financing - causes a given return on assets to be levered up to a higher return on equity. This happens because the firm should earn a higher return on assets than the interest rate on debt. Again this validates the linkage between higher risk and higher return.

Example 14.1: Favourable effect of leverage

Assume companies A and B are identical in all respects except in the way in which they are financed. A is financed entirely by equity, while B is financed 60% by equity and 40% by debt, bearing interest at 8% per year. The tax rate is 27%. The summarised statements of financial position and profit and loss statements for the companies are:

(R millions)	Co. A	Co. B
Total Assets	1,000	1,000
Equity	1,000	600
Debt	-	400
Total equity & debt	1,000	1,000
EBIT	400.0	400.0
Interest	-	-32.0
	400.0	368.0
Tax	-108.0	-99.4
Net income	292.0	268.6
Return on Assets	29.2%	29.2%
Return on Equity	29.2%	44.8%

Both companies earn the same return on assets (NOPAT/Total assets) of 29.2%. The calculation of the return on assets for Co B is indicated as follows:

Net income	Interest	1-tax rate	268.6
After-tax interest charge		32 73%	23.4
Operating income after tax			292.0
Return on assets =		292 / 1,000	29.2%

Otherwise we can simply determine ROA = EBIT(1 – t) /Assets = (400 × 0.73)/1 000.

Through the use of debt, Company B has geared the ROA of 29.2% to a ROE of 44.8% (268.6/600). The after-tax cost of the debt is 5.84%, and this has funded assets which are generating an after-tax return of 29.2%. The net effect is that the shareholders have benefited from the differential in the rates. The shareholders of Co. B have been rewarded for the increased risk they have taken.

Impact on risk

Leverage increases the volatility of the earnings, and the example demonstrated how the return to the shareholders was increased as a result of the leverage. The converse also happens when the return on assets is lower than the interest on the debt. In such circumstances the return on the shareholders' funds is less than the return on the assets. This is known as negative gearing. In severe cases the company may not be able to meet the interest from the earnings, and this puts the firm into very serious difficulties, possibly culminating in bankruptcy. The downturn in airline profitability has highlighted the risks involved in having debt in a recessionary or competitive environment.

Example 14.2: Unfavourable effect of leverage

Let us consider the same companies as in Example 14.1, but in a situation in which business has been difficult and profits are only R30.2m for the year. Page 14-5

The income statements for companies A and B are presented below:

(R millions)	Co. A	Co. B
EBIT	30.2	30.2
Interest	-	-32.0
	30.2	-1.8
Tax	-8.2	-
Net income	22.0	-1.8
Return on Assets	2.20%	2.20%
Return on Equity	2.20%	-0.30%

The return on assets for Co. B is determined as EBIT(1 – t)/Assets = (30.2m × 0.73)/1000m = 2.2%. The return on equity is negative and is determined as -1.8m/600m = -0.30%.

We have indicated that companies such as Woolworths have engaged in share buy-backs in order to move the capital structure and the debt to equity ratio within a target range. We will undertake a more detailed example on the effects of financial leverage on a company's return on equity.

Example 14.3

Tel Ltd is a company that currently has no debt in its capital structure. An investment bank has indicated that the company will be able to create value for its shareholders by paying back funds to its shareholders and borrowing an equivalent amount. The company has an issued share capital of 100 million shares and the share price is currently trading at R6.00 per share. An analyst has indicated that the debt to equity (market value) should be at 100%. This means that the company will borrow R300 million at 10% and use the funds to engage in a share buy-back of 50 million shares at (hopefully) R6.00 per share. The market value of the equity is equal to the book value of the company's net assets. Assume no taxation. What is the effect on the firm's return on equity (ROE) if the company faces five possible levels of operating earnings (A to E)?

Table 14.1 The effect of financial leverage on ROE

Existing Cap. Structure	A	B	C	D	E
EBIT (Rm)	20	40	60	80	100
Interest	0	0	0	0	0
Net Income	20	40	60	80	100
Assets	600	600	600	600	600
Equity	600	600	600	600	600
Debt	0	0	0	0	0
	600	600	600	600	600
Return on Assets	3.3%	6.7%	10.0%	13.3%	16.7%
Return on Equity	3.3%	6.7%	10.0%	13.3%	16.7%
New Cap.Structure	**A**	**B**	**c**	**D**	**E**
EBIT (Rm)	20	40	60	80	100
Interest	-30	-30	-30	-30	-30
Net Income (loss)	-10	10	30	50	70
Assets	600	600	600	600	600
Equity	300	300	300	300	300

Existing Cap. Structure	A	B	C	D	E
Debt	300	300	300	300	300
	600	600	600	600	600
Return on Assets	3.3%	6.7%	10.0%	13.3%	16.7%
Return on Equity	-3.3%	3.3%	10.0%	16.7%	23.3%

Page 14-6 We can see that with financial leverage, the Return on Equity will be more volatile to changes in earnings due to the fixed interest charge. However, the ROE will be greater if the company is able to earn a return on assets that is higher than the interest rate on its borrowings. The steeper line in the following graph depicts the greater sensitivity of ROE to changes in operating earnings.

It is interesting to note Warren Buffett's views on financial leverage and return on equity. He understands that firms can raise their ROE by increasing their debt–equity ratios. Yet he likes to invest in companies with high ROE's without the prop of a high debt–equity ratio. Further, he states that companies with high debt–equity ratios are vulnerable to economic recessions.

Yet we have seen rising ROE's in many companies, partly as a result of companies taking on more debt and also engaging in share buy-backs which reduces a company's equity (in book value terms). The return on market value is another matter. So why do companies try to increase their ROE? Perhaps management wishes to signal to investors that the company may be currently under-rated by the market in relation to its share price and wishes to attract their attention by bolstering the company's ROE.

Truworths reported a ROE of 48% in 2023 due to following a high dividend payout policy and undertaking share buy-backs which has reduced the company's equity. Woolworths has also been restructuring its balance sheet by increasing its debt levels and undertaking share buy-backs in order to raise its ROE.

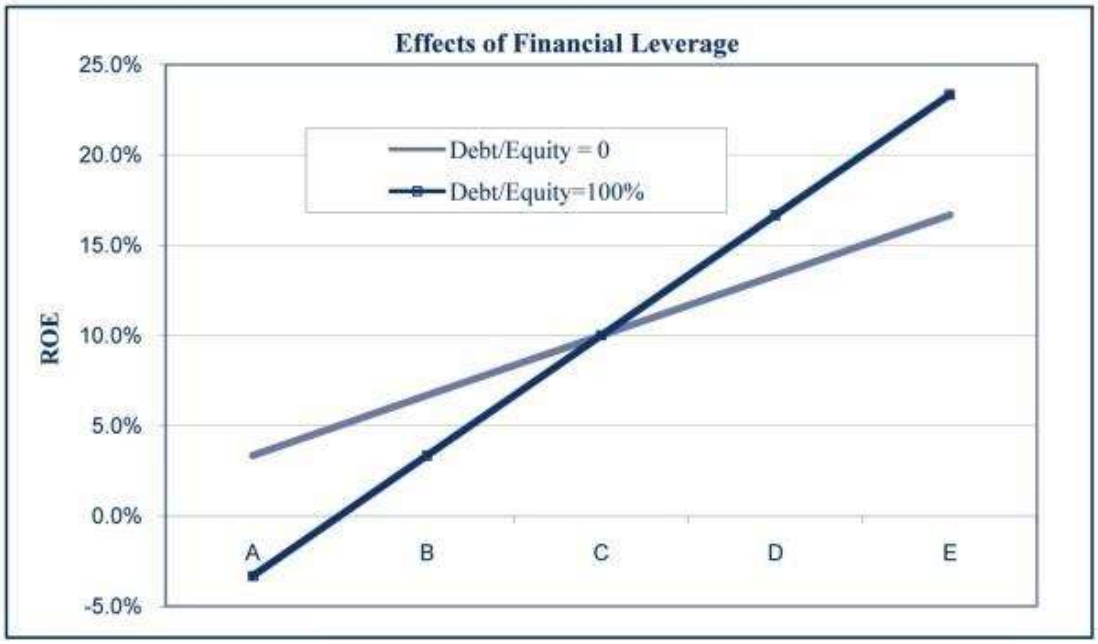


Figure 14.1 Financial leverage and changes to ROE

Let's use another example to focus on the effect that financial leverage can have on earnings. ABC Ltd has assets of R400m, which are financed only with equity. The company is considering borrowing R200m and undertaking a share buy-back or paying a dividend of R200m to reduce equity to R200m so that the assets will be financed 50% by equity and 50% by debt.

Statement of Financial Position	Rm	Adjustments	Rm
Assets	400		400
Equity	400	-200	200
Debt	0	200	200
	400		400

Page 14-7 The interest rate is 10%. The operating earnings of the company in a normal economy will be R48m but can be expected to increase to R60m if the economy experiences an expansion and can be expected to fall to R30m if there is a recession. The percentage change in net earnings will be as follows:

Debt = zero	Recession	Normal	Expansion
EBIT	30	48	60
Interest	-	-	-
Net earnings	30	48	60
Change in earnings	**-37.5%**	**0.0%**	**25.0%**
Debt = 50% of assets	**Recession**	**Normal**	**Expansion**
EBIT	30	48	60
Interest	-20	-20	-20
Net earnings	10	28	40

Debt = zero	Recession	Normal	Expansion
Change in earnings	**-64.3%**	**0.0%**	**42.9%**

The change in earnings is significantly more volatile with debt if there is either an expansion or a recession. Furthermore, if there is a major economic downturn and operating income falls to R15m, then the company may be in danger of failing unless the economy recovers. It is more complicated because the bank may have imposed conditions (loan covenants) and if the company does not meet the conditions, for example by maintaining a minimum interest cover, then the loans may become immediately due and this may push the company into bankruptcy, even though an economic recovery is expected within a few years.

In our examples thus far, we have analysed how financial leverage can significantly affect a company's return on equity (ROE) and earnings. What about other measures of performance such as IRR? Can we increase an investment's IRR by increasing an investment's borrowings? The answer is, yes we can. Assume that you are a manager at a private equity firm, which is investing in a project at a cost of R100m at time zero with the following after-tax operating cash flows. We can compute the IRR to be 16.72%.

	0	1	2	3	4	5
Cash flows	-100.0	40.0	20.0	20.0	20.0	60.0
IRR	16.72%					

Whilst a return of 16.72 % is attractive, in order to attract more funds to your firm, you may wish to bolster the IRR of the investment by undertaking borrowings so that the project will now include debt of R45m at the beginning of the project, with a repayment of the borrowings of R45m at the end of Year 5. We will assume that the after-tax interest rate is 7.2% per year, based on opening balances. The cash flows will be as follows, and due to the inclusion of debt and interest cash flows, the IRR will now increase significantly to 29.83%.

	0	1	2	3	4	5
Cash Flows	-100.00	40.00	20.00	20.00	20.00	60.00
Borrowings	45.00					-45.00
After-tax interest [R45m × 7.2%]		-3.24	-3.24	-3.24	-3.24	-3.24
	-55.00	36.76	16.76	16.76	16.76	11.76

IRR 29.83%

Financial leverage will increase the expected IRR, but this comes with higher risks if operating cash flows fall or interest rates rise. Bain Capital structured its highly

leveraged Page 14-8 investment in Edcon using almost 80% debt and was expecting to earn an IRR of about 25%. However, lower operating cash flows and higher interest costs resulted in significant losses for Edcon and Bain Capital. In 2014, Edcon made a trading profit of R1 299m but incurred a loss of almost R2 449m. Why? Its interest burden on its debt was so large and its trading profit had fallen from R2 290m in 2008 to R1 299m in 2014. The fall in its trading profit was compounded by the high interest burden. By 2016, its trading profit fell further to R987m and the loss (prior to impairments) was R2 829m. In 2017, Edcon was restructured with lenders exchanging debt for equity. In 2020, Edcon was placed in business rescue due to its debt burden and the impact of Covid. Its Edgars division was sold to the Retailibility group, and the JET division was sold to Foschini.

When Edcon was required to refinance part of its debt in 2014, it was required to pay a very high interest rate of JIBAR plus 7% (about 13% in 2014), which indicates the risk of Edcon's capital structure and its high debt ratio. To place this in perspective, in 2014 a newly qualified Chartered Accountant could probably borrow at around 9%. Let's compare Edcon to another leading company, Aspen Pharmacare, that was able to borrow at about 0.9–1.4% above JIBAR in 2013 (about 6–7%) and then raised significant funding of about R7 330m in 2014 at a margin of 1.7–2.1% above JIBAR (about 7–8% in 2014). Aspen's debt ratio then was about 50%. Therefore, we can see that financial leverage may have real economic consequences, not only in terms of increasing the risk to ordinary shareholders, but may also result in a higher interest rate. We will come back to Edcon. In our view, the very high interest rate reflects the potential for financial distress, not only in relation to the shareholders, but also in relation to the banks lending to Edcon. We will also analyse the issue of financial distress later in the chapter. The Glazer family used the borrowing capacity of the Manchester United club, which had been debt free, to effectively take control of the club.

3 OPTIMAL CAPITAL STRUCTURE

The optimal capital structure is *the debt–equity ratio that the company adopts so that its weighted average cost of capital (WACC) is at the lowest point*. There is considerable debate as to whether an optimal capital structure is feasible. Over the years a number of theories have been developed to explain the relevance of capital structure. In 1958, Modigliani and Miller published a seminal paper which presented a

rigorous analysis that capital structure is irrelevant. However, Modigliani and Miller's analysis is subject to certain assumptions. These are:

- The individual can borrow on the same terms as the company can.
- There are no taxes.
- There are no transaction costs.
- There are no costs associated with the financial distress.
- There are no agency costs.

We will now examine the Modigliani–Miller approach and gradually relax the assumptions.

The Modigliani–Miller approach

Modigliani and Miller argued that there is no optimal capital structure because, irrespective of the level of gearing, a firm's weighted-average cost of capital will not change. This occurs because, as the company increases its debt (which is cheaper than equity), it is increasing its risk. The equity holders will require compensation for an increase in risk, and so the cost of equity will rise and offset the relative benefit of the cheaper debt. One of the fundamental principles in this argument is that it is the assets that determine the value of the business and not the way in which those assets are financed. The firm's investment in assets generates the profits, not the way these assets were paid for. This makes sense as smart investments make money, and this is one of the reasons there is so much emphasis on evaluating capital investments, as detailed in Chapter 8. The measurement of this value is based on the *net operating income* Page 14-9 (NOI), which we often call *earnings before interest and taxes* (EBIT). The NOI is capitalised at the WACC which, Modigliani and Miller contend, is a constant, as it is insensitive to changes in capital structure. The value of a firm is consequently independent of its capital structure. Modigliani and Miller establish this by showing that, if this were not the case, arbitrage would take place and drive the prices of the shares to equilibrium. Consider the case of two identical companies: Co. A (financed entirely by equity) and Co. B (financed 40% by debt and 60% by equity). Assume the cost of equity is 20% and the cost of debt 15%.

	Rm	Co. A	Co. B
Net Operating Income		90	90
Interest		-	-30

	Rm	Co. A	Co. B
Available to shareholders		90	60
Value of Equity		450	300
Value of Debt			200
Value of company		450	500

Modigliani and Miller argue that the position shown cannot remain. An investor who holds 10% of the shares in company B can increase the return without increasing the risk. The investor would sell the shares in B for R30m, borrow R20m, and buy 10% of the shares in company A for R45m. The position would then be:

	Rm
Income previously earned: 10% of B's earnings	6.00
Income now earned: 10% of A's earnings	9.00
Less: interest on debt	-3.00
	6.00

The income is R6m in either event, but the investor now still has R5m left over to invest. The risk is unchanged. Whereas previously, shares were held in a company with gearing of 40% of total assets, a personal gearing of 40% of total assets is now in place. Modigliani and Miller's argument is that arbitrage will force the price of B's shares down and the price of A's shares up until equilibrium is reached. The logical deduction is that the cost of equity is moving in sympathy with the capital structure. It follows that the WACC remains constant, irrespective of the quantity of debt financing, and is therefore independent of capital structure. At that point the assumption that value is based on NOI, capitalised at the WACC, which is independent of capital structure, is satisfied. **Page 14-10**

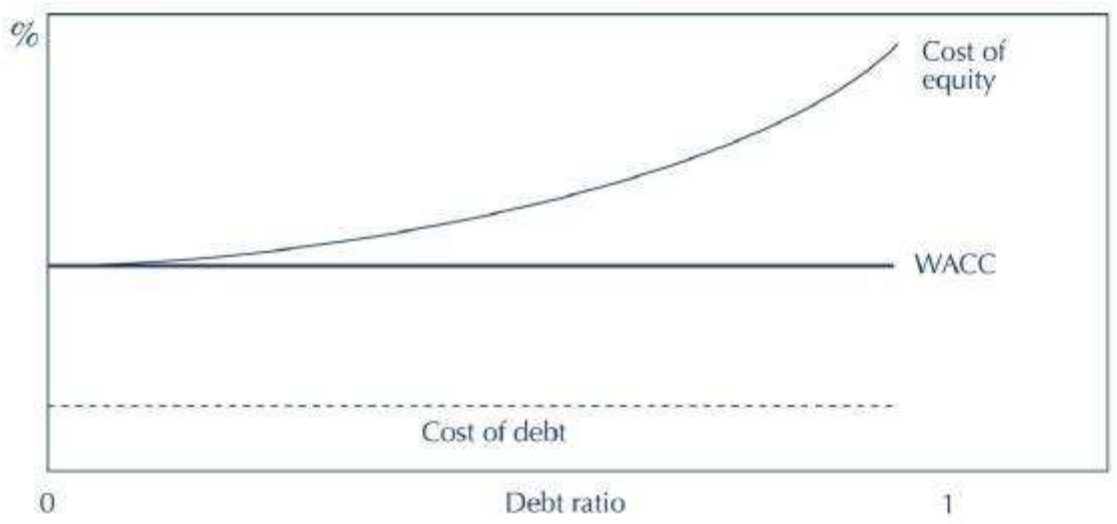


Figure 14.2 The MM approach

Relaxing the assumptions

Relaxing the assumption of no taxes changes the MM position. Since interest on debt is deductible for purposes of taxation, the value of the geared firm will be higher by the amount of tax saved through using debt. The effect is a cost of capital which does decline with increased gearing. This is depicted in Figure 14.3.

Under the MM assumptions, with company taxes, a firm's WACC would decrease continuously as financial leverage increases towards 100% debt finance. Consequently, the value of the firm would increase by the present value of the interest tax shield. This can be expressed in the following equation where V_L is the value of a levered firm and V_U is the value of an unlevered firm.

$$V_L = V_U + (\text{PV of expected interest tax shield}) \qquad \textit{(Formula 14.1)}$$

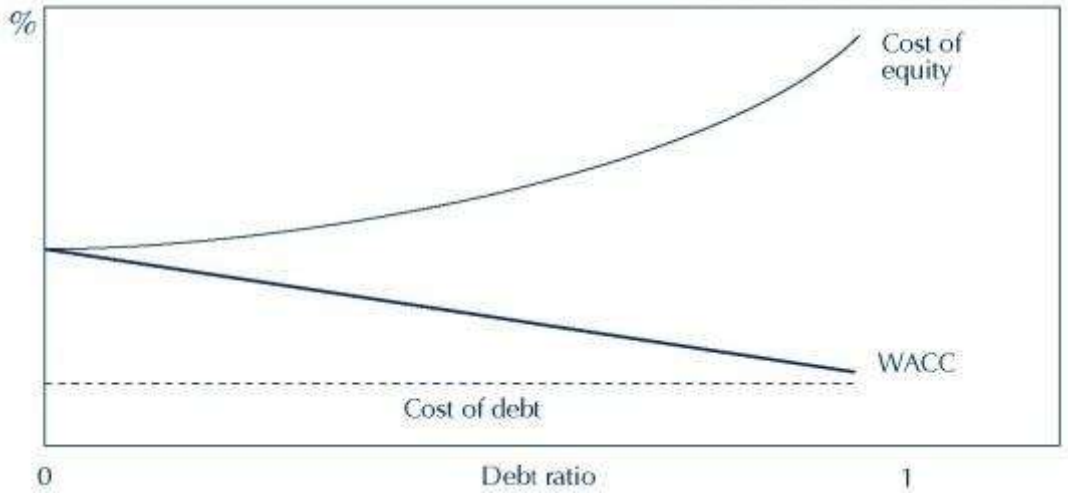


Figure 14.3 The MM with taxes

However, this model does not take into account the fact that, as debt increases, so the risk of potential financial distress increases. It also ignores the fact that EBIT could fall due to a poorly structured balance sheet; it ignores potential agency costs; and it ignores the fact that interest rates are likely to increase at higher levels of debt. Taking these factors into account, the following equation more accurately reflects the situation:

$$V_L = V_U + PV_I - PV_F - PV_{\Delta EBIT} - PV_A - PV_D \qquad \textit{(Formula 14.2)}$$

where: PV_I= PV of expected interest tax shield

PV_F= PV of expected financial distress related costs

$PV_{\Delta EBIT}$ = Reduction in value from lower EBIT

PV_A= Reduction in value from agency costs

PV_D= Reduction in value from increased cost of debt

The above relationship is reflected in Figure 14.4. It can be seen that, initially, the interest tax shield gives rise to an increase in value, but that, from point X, the increase in value declines as expected costs of financial distress and related costs come into play. Point Y is the point where the marginal benefit from any interest tax shield is exactly offset by the increase in marginal cost. This point is the optimal capital structure. Any further increase in debt will result in the marginal disadvantages being greater than the marginal tax shield. While it is possible to estimate the present value of the interest tax shield fairly accurately, the other factors are much more difficult to estimate. For example, to estimate the interest tax shield we only need to estimate whether there will be taxable income, and what the future company tax rate will be. To estimate the expected costs of financial distress is much more difficult as it will depend on countless factors. Page 14-11

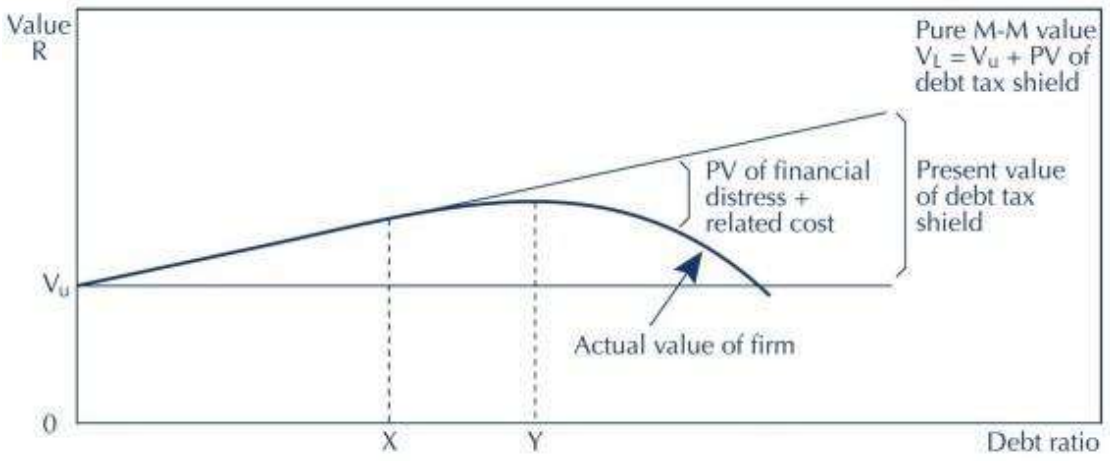


Figure 14.4 The effect of leverage on the value of a firm

While Formula 14.2 does not give us a model that can be readily used in practice to determine the optimal capital structure, it does provide us with a comprehensive conceptual analysis which has given rise to the trade-off theory.

Trade-off theory

A practical approach follows the MM approach with taxes, costs of financial distress, and related costs. While this approach is theoretically consistent with the adjusted MM approach, it is more pragmatic. It recognises that the cost of equity and cost of debt will rise as leverage is increased. This is called the *trade-off theory* because companies are seen to trade off the advantages of debt financing with the costs of taking on increasing levels of financial leverage, such as higher interest rates and potential bankruptcy.

Initially the increase in the costs is considered to be slight, but as the degree of financial leverage increases, the rate of increase in the costs of equity and debt will also increase. As a result the WACC falls slightly initially, then remains reasonably constant for a range of capital structures, and finally increases as the degree of financial risk becomes unacceptable. This relationship is reflected in Figure 14.5.

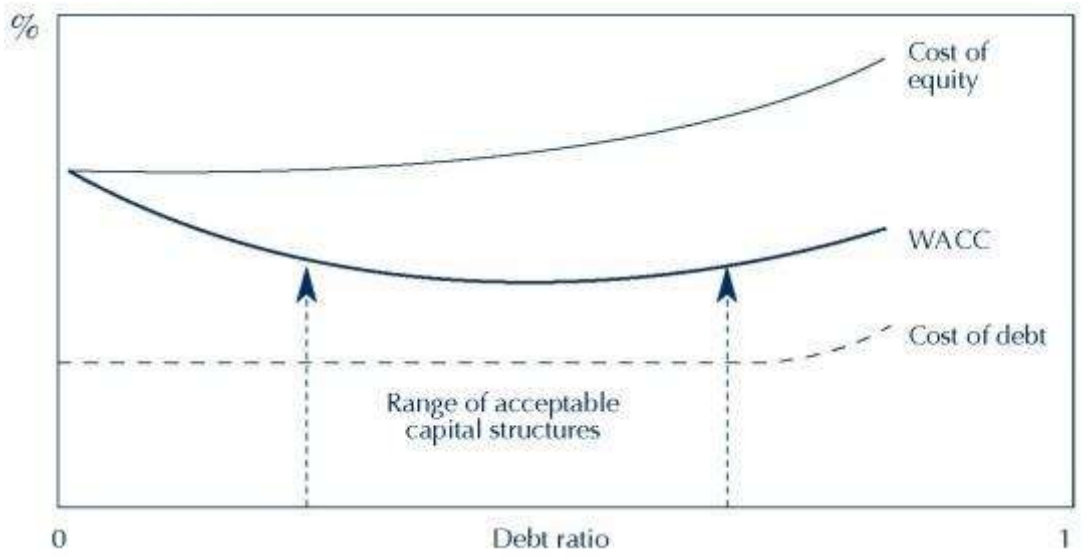


Figure 14.5 A practical approach

The significance of this approach is that it considers a whole range of capital structures to be optimal. Thus the selection of an appropriate capital structure is made easier, and the implementation less sensitive. As assets tend to be financed by either debt or equity, the capital structure will change marginally each time an asset is acquired. This is of no concern to the financial manager, however, so long as the capital structure is kept within the relevant range. Page 14-12

In the real world lenders such as banks will not offer firms loans unless there is sufficient security from the company's assets or its cash flows. A new company in the biotech sector will find it difficult to obtain debt financing. It will have to rely on equity financing. However, a company that has been operating for many years and is generating positive cash flows will find it easier to obtain debt financing. The level of borrowings will be determined by the risk of the company's operations, its assets and the nature of the industry sector. Traditionally, utilities were able to generate a high level of debt due to the stability of their cash flows and their monopoly position in the market. As energy, water and other utility sectors become increasingly competitive, their debt to equity ratios will decline.

Companies have varying debt–equity ratios and therefore the trade-off theory may be able to explain what we see in the real world. Companies do limit their use of debt in order to reduce the potential bankruptcy costs. What do these bankruptcy costs include? Firstly, these costs relate to the legal and accounting costs of winding up but also include costs such as the sale of assets at distressed prices due to the specialised nature of a company's assets. Further, there are significant *indirect* costs arising from the threat of bankruptcy. These include the following:

- Customers leave as they consider that the company will not be able to provide servicing and spare parts in the future, nor will it be able to support product warranties. Would you purchase a motor vehicle if you thought that the manufacturer might be bankrupt within a year? You would worry about future servicing, availability of parts and that the warranty may have no value. Let's face it, you would think twice about flying with an airline that was in a precarious financial state. Of course, companies may reduce prices to support sales, but this will result in a fall in margins.
- Key employees will leave or be demoralised by the company's weak prospects.
- Banks may impose higher interest rates or may request loans to be repaid. The company may have contravened loan covenants thereby granting banks the right to recall longterm loans.
- Investment opportunities that are profitable and critical for the company's long-term future are not able to be given the go-ahead as the company has limited financing options.
- The focus on keeping the company afloat will divert management time and resources away from such critical functions as human resources, production, marketing and product quality.
- Companies will tend to reduce research and development, advertising, and maintenance which may have longer-term effects on future profitability.
- Competitors will take increasingly aggressive actions to reduce the ability of the company to compete. For example, a competitor may engage in a price war as it knows that the company will not be able to survive, thereby resulting in one less competitor in the sector.
- Suppliers may withhold supply or demand prepayment.

The maintenance of a low debt–equity ratio provides assurance to customers, employees and suppliers that the company will continue even if it faces some difficult trading conditions in the future. What does Warren Buffett think about the use of debt? In setting out his acquisition criteria for investing in companies, Buffett states that he is looking for "businesses earning good returns on equity while employing little or no debt".

Let's come back to the trade-off theory. If firms limit their use of debt in order to balance the advantages of debt with the costs of bankruptcy, then we would expect to see very profitable firms make increased use of debt. Yet this is not always true. Many large and very profitable companies use less debt than indicated by the trade-off theory. This suggests that although the trade-off theory offers us a useful insight into capital structure decisions, it is not the complete picture. The pecking order and signalling theories may help explain why. Page 14-13

Pecking order and signalling theories

Pecking order theory assumes there is no target capital structure. Rather, finance will be raised according to a preferred hierarchy. In general this hierarchy suggests that financing investments from retained earnings is the most preferred alternative. Additional funds will be raised, in order of preference, from straight debt, to finally, the issue of new equity. This hierarchy accommodates two attributes that managers value: flexibility and control. Retained earnings preserve the existing control of the firm, because debt holders may impose restrictive covenants, and new equity dilutes existing control. As far as flexibility is concerned, debt reduces flexibility in raising future finance. Information asymmetry is also a factor. As internal managers are more likely to have better information about the firm than external investors, they would know if the firm's financial instruments are mis-priced. Management would be reluctant to issue equity, which is under-priced, consequently an equity issue may be interpreted as a signal that the equity is overpriced.

The order in which funds would be issued is thus:

1. Internally generated funds – retained earnings
2. Debt
3. Convertible debt and preference shares
4. Equity

The pecking order theory states that management will always use the lowest-cost financing alternative. Management will use internally generated financing such as retained earnings before making use of external financing. If external financing is required, then a company will firstly prefer to use debt as it is of lower cost than the issue of new equity. Companies will issue new equity only when forced by circumstances or when the company is not able to raise any further loan finance.

In line with the pecking order theory, we would expect to see low debt–equity ratios for firms that are generating strong cash flows but have limited growth opportunities.

The use of equity financing acts as a signal to the market that management perceives the share price to be over-valued. If the company has significant profit opportunities, then it will endeavour not to issue shares and will try to rather use retained earnings or debt if this is not possible. A company that is actively buying its own shares is signalling to the market that its share price is understated. The issue of debt indicates that the company has excellent prospects and the equity holders wish to retain the benefits from any future investment opportunities.

Companies will wish to avoid issuing new equity to finance investment opportunities that may arise due to the negative signalling effects of such an issue. Therefore a company will maintain a lower than expected debt–equity ratio indicated by the trade-off theory so that it is able to quickly take advantage of any investment opportunities without having to issue new equity. The company will maintain spare borrowing capacity or surplus cash balances so that it is able to quickly take advantage of investment opportunities. The company is said to own a "war chest" for future investments. Further, the existence of spare borrowing capacity means that the company is able to undertake investments without risking making such information available to the capital markets. Companies often maintain spare production capacity in order to immediately take advantage of increased market demand. It is to be expected that companies would also maintain spare borrowing capacity. This helps explain what we see in the real world.

Debt financing, free cash flow and conflicts between management and shareholders

We saw in Chapter 1 that there may be agency problems between management and shareholders. Any potential conflict between management and the firm's shareholders is accentuated when the firm has lots of free cash flow. Therefore, shareholders may take on more debt in order to discipline management so that

management avoids any waste and unnecessary capital expenditure. The use of debt forces management to maximise returns and minimise the use of capital. If you are walking a tight rope you tend to be highly focused on the task at hand. The use of debt financing has an explicit cost which equity does not have. One of the main reasons for using EVA is to ensure that the cost of using equity capital is taken into account in measuring financial performance. However, the disadvantage of using a high debt–equity ratio to influence management action is that sometimes there are economic slowdowns that are not the fault of management and the future of the firm is placed at risk owing to the high debt levels. The use of leveraged buyouts fundamentally changed the American economic landscape and made firms lean and highly focused on their core competencies. Yet the use of EVA and aligning management incentives with shareholder wealth maximisation may be a smarter thing to do.

4 AGENCY COSTS AND INVERTED INCENTIVES: CONFLICTS BETWEEN SHAREHOLDERS AND BONDHOLDERS

Whilst it is true that free cash flow may create agency issues, the reality is that if debt leads to financial distress, then this will accentuate the conflicts of interest between shareholders and bondholders. Why is this? In Chapter 1 we set out the possible conflicts between shareholders and bondholders. We need to understand that the incentives when firms are close to the edge move away from maximising NPV and towards shareholders and bondholders looking after their own self-interests. What actions would shareholders tend to take when highly levered or in financial distress?

Investing in high-risk projects

When a firm is close to failing, shareholders may be tempted to *bet the farm* on one last investment. The firm is going to go down anyway and the losers will tend to be the bondholders. How does this work? We will assume that XYZ Ltd has assets of R100m which is financed by loans of R60m and equity of R40m. The recession hits the company hard and the value of the company's assets falls to R50m. This means that there is no equity value[1] and the bonds have a value of R50m. The bondholders are still owed R60m but the value of their bonds have fallen to R50m. Let's assume that the company is able to now undertake two projects with the following probabilities and possible NPVs:

Projects	Probability	NPV	Expected NPV
Project A	50%	10	5.0
Outcome 1			
Outcome 2	50%	20	10.0
			15.0
Project B	75%	0	0.0
Outcome 1			
Outcome 2	25%	50	12.5
			12.5

If we select Project A, then the expected NPV is R15m whilst if the firm selects Project B, then the NPV would be R12.5m. On the basis of NPV, the firm should select Project A. Yet, the shareholders may wish to select Project B. Why? The shareholders will recover a Page 14-15 value of only R10m (50 1 20 – 60) once the bondholders are paid if Outcome 2 applies for Project A. Outcome 1 will mean zero for the shareholders as all the value will go to the bondholders. In Project B, the shareholders have a 25% chance of recovering R40m (50 1 50 – 60) after the debt holders are paid. This means that 25% of R40m (R50m - R10m payable to the bondholders) is better than 50% of R10m (R20m - R10m payable to bondholders). The bondholders would prefer the firm to take Project A because there is a 100% chance that they will recover the loss of R10m whilst for Project B, there is only a 25% that they will recover R10m. A capital structure that is highly levered which results in financial distress will mean higher agency costs. Debt holders will be desperate to protect the value of their assets when financial distress hits the firm as they understand the incentives that shareholders have. When shareholders have nothing, they have nothing to lose and may take high-risk bets as they then own most of the upside and the bondholders own all the downside. The profits go to the shareholders and the losses go to the bondholders.

Running off with the money

If the firm is teetering towards collapse, then the shareholders may bolster their dividend payouts, make distributions to themselves, spend lavishly, and make asset sales which will often place the firm in further difficulties. Shareholders will be incentivised to increase their own returns at the cost to the bondholders. This is like the proverbial rats abandoning the sinking ship with all the (floating) possessions they can muster so that the bondholders are left with little to recover after bankruptcy. The debt holders will try and prevent this happening but are not always successful. The

Companies Act also imposes solvency and liquidity requirements on the board of directors which may make directors personally liable for reckless actions so that they do have something to lose. The Companies Act's solvency and liquidity requirements are set out in greater detail in Chapter 16.

No further investment by shareholders

Shareholders will not be incentivised to make further investments into a firm that is highly levered and close to bankruptcy. Why not? An example will illustrate this. Let's go back to XYZ Ltd which has assets with a book value of R100m but a market value of R50m. The bondholders are owed R60m but the market value of their bonds is R50m. Assume the company is able to invest in a project with an NPV of R10m but the shareholders would need to invest a further R20m in the company to achieve this positive NPV. The shareholders will end up with R20m (being what they invested) and all the NPV of R10m will go to the bondholders so that the value of their loans is increased to reflect their book value of R60m. There is no incentive for the shareholders to invest any further amounts in the company because all the gains go to the bondholders.

Playing for time

When a firm experiences financial distress, bondholders would like to realise the assets as quickly as possible to reduce any further losses and reduction in asset values. For shareholders, the opposite applies – they would like to delay bankruptcy for as long as possible as something may arise suddenly that will increase the value of equity. For example, assume an exporter is experiencing losses due to the strong Rand and is facing bankruptcy. The shareholders may like to hold out as long as possible – who knows – the rand could collapse suddenly saving the interests of the shareholders. For the debt holders, any further delays in realising assets will increase losses to the bondholders. How can the shareholders and management do this? Delay tactics include creative accounting such as reducing depreciation on assets, reducing maintenance, reducing research and development and reducing advertising and marketing costs. This will make operating performance seem better than it really is. Watch out for changes in accounting policies and watch those ratios!

Changing the capital structure of the firm

The decision to move to a high debt–equity ratio and increase debt in the capital structure of a firm will mean a transfer of value from the bondholders to the shareholders. The risk to all bond holders will increase and this will therefore result in a loss in the value of bonds issued to date whilst this will represent a transfer of value to the shareholders.

Conflicts between shareholders and bondholders – the case of Edcon

Do these conflicts occur in the real world? In the Edcon annual financial statements, which are produced mainly for the holders of its notes (debt/bond holders), the group stated the following to its note holders/bond holders[2]:

> We are indirectly owned and controlled by investment funds advised by Bain Capital. The interests of our equity holders may not in all cases be aligned with your interests. For example, if we encounter financial difficulties or are unable to pay our debts as they mature, the interests of our equity holders might conflict with those of the holders of our outstanding notes or with those of other stakeholders. In addition, our equity holders may have an interest in pursuing acquisitions, divestitures, financings or other transactions that, in their judgment, could enhance their equity investments, even though such transactions might involve risks to holders of the notes.

The use of loan covenants to manage shareholder and bondholder conflicts

What can debt holders do to protect against shareholder actions when firms are highly levered or in financial distress? The use of loan and financial covenants is very valuable for a debt holder. For example, if the firm triggers a break in a loan covenant, long-term loans can become callable which enables the debt holder to protect asset values by quickly realising these assets or the lender can reprice the risk by raising interest rates and imposing other conditions on the firm. The following are some of the covenants that can be imposed:

- the company may not issue additional debt without the permission of the current bondholders;
- any mergers or acquisitions may not be permitted unless authorised by the current bondholders;
- limits may be imposed on share buybacks and dividend payments to shareholders;

- the firm is not allowed to dispose its major assets unless authorised by the current bondholders;
- the firm is not permitted to offer guarantees or pledge its assets to other firms; and
- a firm may need to comply with minimum working capital ratios, interest coverage ratios, debt–equity ratios and maintain a certain value of shareholders' equity investment in the business.

Covenants may also be used to limit a firm's use of derivatives such as interest rate swaps, forwards, futures and options unless such use reduces the risks to the debt holders. In fact, banks may force firms to sell forward in order to fix future revenues. This occurred in the gold mining sector when bank lending to some mining firms was subject to these firms undertaking forward sales of future gold production. At the time, this was not a good idea for the shareholders but it protected the debt holders. Page 14-17

FROM THE REAL WORLD ... LOAN COVENANTS AT ASPEN AND GOLDFIELDS

Aspen is a leading world pharmaceutical group which is based in South Africa. Aspen disclosed the following in its 2022 annual financial statements:

'In terms of the Group's funding arrangements with its lenders, the Group was subject to the following financial covenants in the year:

- Debt (net of cash and cash equivalents) may not exceed this multiple of earnings before interest, tax, depreciation, non-recurring items and amortisation (EBITDA) 3.5
- EBITDA must not be below this multiple of net finance charges 3.5

We can see that lenders use covenants to manage the risk of lending to Aspen. From Aspen's perspective, loan covenants will reduce the interest rate on the loans from the lenders.

Business Report (quoting *Bloomberg News* on 1 September 2014) referred to **Gold Fields'** strategy of reducing debt over the next three years in line with expectations that the gold price may become increasingly volatile. This was due to the risk that a lower or more volatile gold price may result in Gold Fields transgressing its loan covenants. Gold Fields' banks limit its net debt to 2.5 times EBITDA. The CFO, Paul Schmidt, stated that although Gold Fields' net debt was at 1.5 times EBITDA at the time, this could easily rise to 2.5 times EBITDA if there were problems within the sector. *Business Report* then quoted him saying that if this happened, then the banks would be all over them. "As with some of our peers, they force you to do stupid things to try and fix up your financing, either doing rights offers or ridiculously priced bonds". Therefore, it is true to say that in line with an uncertain operating environment

and greater volatility in the pricing of gold, the group is highly focused on reducing its level of financial risk over the next few years to about 1 times EBITDA. This indicates the power of loan covenants in affecting strategic business decisions. So, what has happened to Gold Fields' loan covenants more recently?

In the 2022 annual financial statements, Gold Fields stated that:

> The Group's long-term target is a ratio of net debt to adjusted EBITDA of one times or lower. The bank covenants on external borrowings require a net debt to adjusted EBITDA ratio of 3.5 or below and the ratio is measured based on amounts in United States Dollar.

Gold Fields in 2022 indicated debt to adjusted EBITDA of only 0.29, significantly within its loan covenant parameter.

5 THE IMPACT OF INFLATION

When the impact of the interest tax shields on the cost of debt was considered, it was assumed that there was no inflation. As tax shields are based on the nominal cost rather than the real cost of debt, the real after-tax cost of debt decreases as inflation increases. This is best explained through the use of an example.

Example 14.4: The real cost of borrowing after tax

A company borrows R100m for one year. The current inflation rate is 10% and the lender requires a real rate of return of 10%. Assume the company tax rate is 30%. If we consider what the company is required to repay at the end of the year, it would amount to R121m, being capital of R110m plus interest at 10%, amounting to R11m. Therefore the nominal interest rate is 21% when the real interest rate is 10%, and the expected inflation rate is also 10%. This relationship can be expressed in the following formula:

$$I_n = (I_r + R) + (I_r \times R) \qquad \textit{(Formula 14.3)}$$

where: I_n = nominal interest rate

I_r = real interest rate and

R = rate of inflation

Page 14-18 Returning to the example, we now need to consider the position after tax. For tax purposes the company will be allowed to deduct the nominal interest of R21m, generating a tax shield of R6.3m. However, the real interest amounts to only R11m, and so the real after-tax cost of borrowing R110m is R4.7m. Therefore the real after-

tax cost of borrowing is only 4.27%. The real after-tax cost of debt can be calculated using the following formula:

$$K_{dr} = \frac{(1 + K_{dn})}{1 + R} - 1 \qquad \textit{(Formula 14.4)}$$

where: K_d = real after-tax cost of debt

K_{dn} = nominal after-tax cost of debt

R = rate of inflation

$$K_{dr} = \frac{1 + 0.21\,(1 - 0.3)}{1 + 0.1} - 1$$

$$= 0.0427$$

The significance of the above relationship is that *the higher the rate of inflation, the lower the real after-tax cost of debt*. Inflation may therefore impact on the optimal capital structure by encouraging a higher degree of financial leverage. However, inflation may also increase the level of financial risk facing a company.

6 THE NEED FOR FLEXIBILITY

It is imperative that the financial manager does not view financing decisions with the rigidity that is implied by the theoretical models. The task should be approached with a degree of flexibility with an eye on market conditions and ability to pay.

Target capital structure

If the company does accept that it has not reached its optimal capital structure, then it will strive to reach a target capital structure that it considers optimal. This does not necessarily mean that the company will permanently be at exactly that target. Rather, it means that, over a period of time, it will be striving towards that position. When it deviates for particular reasons it will subsequently be trying to redress the position and return to the target. In practice companies are always re-balancing their balance sheets with a combination of so called capital initiatives. These include share buybacks, the issuing of debt and special dividends.

Short-term deviation from target

A company is often faced with opportunities arising from market inefficiencies. It can take advantage of these inefficiencies. For example, the opportunity may arise to raise longterm debt at favourable interest rates. In such a situation it would seem foolhardy to forego this opportunity simply because it would disturb the capital structure. The

company would utilise this opportunity to raise the debt and to redress the situation over the next few years in such a way that the target is once again approached. It must be borne in mind that financial management is a practical exercise and that financial theories are there to guide and assist in the financial management process. The theories are not intended to be rigidly applied, since they are invariably based on simplifying assumptions. Although they provide insight and an understanding of the mechanisms of certain financial decisions, there are nevertheless times when certain tactical manoeuvres are useful in achieving longer-term strategies that may seem inconsistent with theory. There may be other reasons for a more permanent deviation from what would be considered to be a target capital structure. This relates to financial flexibility and market timing. Page 14-19

Financial flexibility

A company may decide to maintain borrowing capacity and have a very low debt–equity ratio, either to quickly take advantage of opportunities that arise in the future or to be able to survive a cyclical downturn. For example, during upswings, a construction company may reduce debt so that it is well below its longer-term target debt–equity ratio. When there is an economic downturn, the company will be able to utilise its borrowing capacity in order to give it time and the required financing to restructure its operations, downsize and refocus its operations on profitable divisions. While a company's Statement of Financial Position may indicate a debt/equity ratio that is lower than optimal, we need to consider that the management of its debt may be optimal over the economic cycle.

Warren Buffett holds onto significant cash resources and while he would like to invest this cash, this also offers him the flexibility to undertake an acquisition very quickly without the requirement to borrow or issue new shares. **Aspen Pharmacare** was able to invest in GlaxoSmithKline brands because it had available spare borrowing capacity.

Tiger Brands states the following in its 2022 annual financial statements:

> The company and group manage its capital structure, calculated as equity plus net debt, and makes adjustments to it, in light of changes in economic conditions. To maintain or adjust the capital structure, the company and group may adjust the dividend payment to shareholders, return capital to shareholders, issue new shares or increase or decrease levels of debt. The company and group monitor capital using a gearing ratio, which is net debt divided by total equity. The company and group target a long-term gearing ratio of 20% to

30%, except when major investments are made where this target may be exceeded. During the year, the group embarked on a share buy-back programme which is in line with the policy to maintain the gearing ratio.

In 2023, in order to manage its high level of borrowings and avoid financial distress, **Nampak** planned to make a rights issue of R1.0 bn later in 2023. Yet its market capitalisation was only R482m on 30 June 2023. The effect is to ensure significant equity dilution if shareholders do not follow their rights. However, Nampak's debt to book equity was 348% in 2022 and in relation to the debt to market capitalisation (market value of equity) ratio, it was significantly worse as Nampak was trading at a price to book ratio of only 0.21. Its high level of debt was impacting on the viability of the firm and severely limited its financial flexibility. Just 10 years ago (June 2013), Nampak's share price was close to R33.00 and on 30 June 2023, it was only R0.75. Nampak's total debt of over R8bn was an important factor that led to the destruction of shareholder value.[3]

A significant increase in debt at **PPC**, a major cement producer, led to a ratings downgrade of its debt to below investment grade. This made its bonds immediately payable. PPC did not have financial flexibility and was required to make asset sales and make a rights issue at R4.00 per share, when the share price then was close to R9.00, in order to survive.

Market timing theory

In terms of the market timing theory, firms will adjust the capital structure in accordance with market conditions. For example, if share prices are trading at high levels and high P/E ratios, then a firm will tend to issue equity to take advantage of high share prices relative to book values and recent market values. In contrast, if interest rates are very low in relation Page 14-20 to historical rates, then firms will often borrow or issue bonds to take advantage of the low interest rates. Bidvest's statement in its 2022 annual financial statements below is in line with the market-timing theory that capital structure depends on market conditions, rather than on other reasons indicated by the other theories of capital structure such as the pecking order and trade-off theories. This also means that a firm's capital structure will simply reflect the cumulative effects of the firm timing the equity markets and bond markets. However, Bidvest seems to balance its use of debt and equity.

Bidvest states the following in its 2022 annual financial statements:

In the early days of this Group, acquisition activity was generally funded via the raising of equity capital. However, over the past five years, far more favourable credit markets have enabled the use of debt as a far more effective tool of capital. The current credit markets have been extremely volatile, increasing the cost of debt in the weighted average cost of capital thereby enabling a potential return to tapping the equity markets to fund future growth.

7 DEBT AND TAX SHIELDS

In South Africa, a company is permitted to deduct interest incurred if such interest is in the production of income. When we compare debt financing with equity financing, this forms a crucial difference between these sources of financing. If we finance the acquisition of assets with debt, we are able to deduct interest for tax purposes. If we finance the acquisition of assets with ordinary equity or preference shares, we may not deduct dividends from the taxable income of the firm. Assume that XYZ Ltd invests R1 000 million in operating assets and management finances this with a loan. If the firm pays interest of R100m, then this will result in a tax deduction of R100m and a tax saving of R27m per year. Surely this has value. If the financing stays in place or is refinanced indefinitely, then we will receive a tax saving each year in the future.

The firm is expected to obtain the tax saving as long as its income (before interest) exceeds the interest charge. This means that the risk of the tax saving is expected to be less than the risk of the firm and so the use of WACC as the discount rate is not appropriate. The risk of the tax saving is often assumed to be equal to the underlying loan and the interest rate is often used as the discount rate to value such tax shields[4]. Investors are currently requiring a return of 10%.

$$\text{Value of tax shield} = \text{R27m}/0.10 = \text{R270m}$$

Let's restate this by using a simple formula, which uses the interest rate (r), the loan amount (D) and the marginal corporate tax rate (T) to determine the value of the interest tax shields:

$$\text{Value of tax shield} = (T \times rD)/r = TD$$

The interest rate will fall away from the formula so that we end up with the tax rate multiplied by the debt amount (TD).

$$\text{Value of tax shield} = 0.27 \times \text{R1 000m} = \text{R270m}$$

Let's come back to MM. If MM applies except for taxation, then we can determine the value of the firm as follows:

Value of Company = value of firm if all-equity financed + value of tax shield

Page 14-21 In this case, the market value of the all-equity financed firm would be equal to the value of the assets, which is R1 000m. The value of the tax shield is R270m resulting in a total value of R1 270m.

The value of the tax shield is equal to the tax rate multiplied by the loan amount and this is often presented as the way to calculate the value of tax shields. However, we are assuming a permanent level of debt financing. Normally, debt levels will increase over time if the company maintains a fixed debt–equity ratio. We then need to discount the expected tax shields for each year and this will increase the value of the tax shields in relation to the formula that the value of the tax shield is equal to TD. We can apply a growing perpetuity formula to determine the value of the tax shields.

If the tax shields are expected to apply only for a number of years, then we need to discount the specific tax shields for each year. For example, in a leveraged buyout, we would expect the tax shields to fall over time as the company moves to a lower and normalised debt–equity ratio over time.

Let's return to our example. Assume that the operating return on assets is 15%, so that the firm is earning R150 million in operating income. As the interest rate is 10%, this will result in net income before tax of R50m (R150m – R100m). The tax shield has value. Let's assume however, that the economy enters recession and the company is only able to achieve an operating return of 6% on assets. The company is making a loss of R40m (R60m – R100m) and this means that the tax shield on this loss is either partially lost or deferred. Therefore, for firms with high debt–equity ratios the deductibility of interest may not always result in tax shields. It depends on the firm's operating returns and level of debt. In most cases, the tax shields are reasonably certain but times can change.

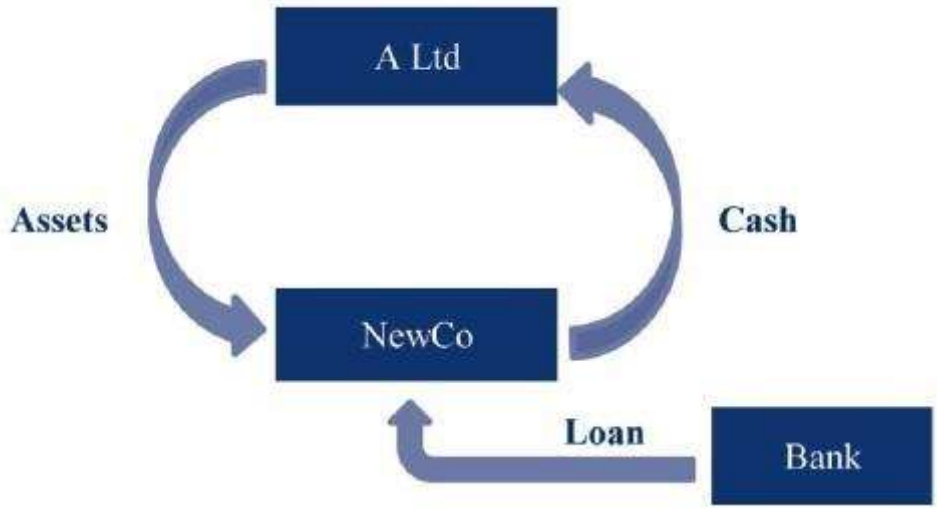
If interest tax shields add value, then why are firms not structured only with debt? There is an important agency relationship that arises and we explained before that the costs of financial distress will reduce the amount of debt that a firm will often use in practice. Also, lenders will not wish to lend beyond a reasonable level – otherwise the banks will end up with all the risk and none of the upside of equity. Interest rates become punitive as we increase the level of leverage. However, there may be security offered from outside the corporate entity which enables the firm to increase its level of loan financing from the banks at a reasonable interest rate[5].

We will now evaluate the impact that a change in capital structure arising from a corporate restructuring can have on the value of equity. Further, to understand the relative value of interest tax shields, we need to consider personal taxes, which we will do later in the chapter.

Example 14.5

Page 14-22 A Ltd has assets with a market value of R1 000m which are currently financed 100% with ordinary equity. The company is selling the business as a going concern to NewCo and A Ltd owns 100% of the shares in NewCo. This means that the operating company will be placed within NewCo. A bank will lend R900m to NewCo resulting in a highly levered company in order to maximise the interest tax shields. A Ltd will provide some security to the bank to ensure that interest rates are set at a reasonable level. The interest rate is 12%. We will assume that the tax consequences of the transfer of the business assets to NewCo are insignificant and we will focus only on the tax shields arising from the levered structure.

The transaction may be depicted as follows:



NewCo will receive the bank loan of R900m and pay this over to A Ltd in consideration of the assets acquired from A Ltd. We will assume that A Ltd will invest equity financing of R100m in NewCo. The financial statements of the operating company before and after the transaction will look as follows:

Table 14.2 Corporate restructuring with financial leverage

	A Ltd	NewCo
Statement of Financial Position	Rm	Rm
Non-current assets	700.0	700.0
Other assets		

Net working capital	300.0	300.0
	1000.0	1000.0
Equity	1000.0	100.0
Long term loans	0.0	900.0
	1000.0	1000.0
Statement of Comprehensive Income		
Operating income (EBIT)	200.0	200.0
Less: Interest	0.0	-108.0
	200.0	92.0
Tax	27%-54.0	-24.8
	146.0	67.2
ROE	14.6%	67.2%
ROA	20.0%	20.0%

Firstly, we can see that the before-tax operating return on assets of 20% (200/1 000) has not changed but the return on equity (ROE) has increased significantly from 14.6% (146/1000) to 67.2% (67.2/100).

If MM applies except for taxation, then we can determine the value of the firm as follows:

$$\text{Value of Company} = \text{value of firm if all-equity financed} + \text{value of tax shield}$$

In this case the value of the tax shield is equal to R243m, which is determined as follows:

$$\text{Value of tax shield}^{6} = \text{R900m} \times 0.27 = \text{R243m}$$

Page 14-23 The total value in the group is then R900m + R100m + R243m = R1 243m[7] We will now determine the likely impact on the WACC as a company changes its financial leverage.

8 FINANCIAL LEVERAGE AND A FIRM'S WEIGHTED-AVERAGE COST OF CAPITAL

In Chapter 7, we calculated the impact that a change in financial leverage is expected to have on a company's cost of equity and weighted-average cost of capital. We employed the Hamada formula to unlever and relever betas to reflect changes in financial leverage. We also used this to adjust for project risk. For the sake of completeness, we will do a simple example to illustrate the effect the change in financial leverage will have on the firm's beta and its cost of equity.

Example 14.6

MU Ltd is a company with an equity beta of 1.10. The yield on long-term government bonds is 8% and the market risk premium is 5%. The firm's current debt–equity ratio is 50%. The company's cost of debt is 10% and the tax rate is 27%. The company wishes to change its capital structure and increase its debt–equity ratio to 150%. What impact will this have on the firm's WACC?

Firstly, we will unlever and relever the current beta as follows:

Unlevering the current beta	c	**d**	e	f = c/e
	Beta	**Debt/Equity**	**[1+(1-T)D/E]**	**Unlevered Beta**
$B_U = B_L/[1+(1-T)D/E]$	1.10	50.00%	1.365	0.806
	f	g	h	i = f×h
Relevering beta to the new D/E ratio	**Unlevered Beta**	**Debt/Equity**	**[1+(1-T)D/E]**	**Relevered beta**
$B_L = B_U \times [1+(1-T)D/E]$	0.806	150.00%	2.095	1.69

We can now use the re-levered beta and altered capital structure to determine the company's revised cost of capital with a debt–equity ratio of 50% and a debt–equity ratio of 150%.

Table 14.3 The weighted-average cost of capital with varying betas and debt–equity ratios

Weighted Average Cost of Capital			D/E = 50%			D/E = 150%
Risk-free rate (long-term govt bond)			8.0%			8.0%
Market Risk Premium		0.050			0.050	
Beta	x	1.10	5.5%	x	1.69	8.5%
Cost of Equity			13.5%			16.5%
	Cost of debt	1-tax		Cost of debt	1-tax	
Cost of Debt (after tax)	10.0%	0.73	7.3%	10.0%	0.73	7.3%
			WACC			WACC
Cost of Equity	13.5%	0.67	9.0%	16.5%	0.40	6.6%
Cost of Debt (after tax)	7.3%	0.33	2.4%	7.3%	0.60	4.4%
			11.4%			**11.0%**
Target Debt-Equity		50%			150%	

Page 14-24 As we can see, although we have increased the financial leverage from a debt–equity ratio of 50% to 150%, the cost of capital has fallen only by 0.4% as the greater use of after-tax cost of debt is offset by a higher cost of equity. We have also assumed that lenders will continue to lend at 10% and this is unlikely as the company increases its financial leverage[8].

De Wet (2006)[9] undertakes the determination of cost of capital and capital structure for three South African companies and increases the cost of debt as financial leverage increases so that there is an optimal capital structure that minimises a company's cost of capital. Of course, even a 1% reduction in a company's WACC can have a material impact on value. Private equity funds often set out the advantages of using lots of debt to finance a leveraged buyout.

9 PERSONAL TAXES

In our study of finance so far, we have avoided the issue of personal taxes and we have focused only on corporate taxation. Individuals in South Africa can pay up to a marginal tax rate of 45% but dividends are taxed separately. The implementation of a dividend withholding tax system means that shareholders will pay a final tax on dividends received of 20%. However, we are going to change the facts for now to analyse whether debt retains its advantage after we account for personal taxes. Let's go back to our example of A Ltd and NewCo.

		A Ltd	NewCo
Statement of Comprehensive Income		Rm	Rm
Operating income (EBIT)		200.0	200.0
Less: Interest (payable to bondholders)		0.0	-108.0
		200.0	92.0
Tax	27%	-54.0	-24.8
Net income (due to shareholders)		146.0	67.2
Total income to shareholders & bondholders		146.0	175.2 [108+67.2]

As we can see the total income to shareholders and bondholders is increased through the use of debt [(108 + 67.2) > 146)]. We will now take personal taxes into account. Firstly, we will assume that interest income and dividend income are both taxed at a personal tax rate of 41%. The total income to the shareholders and bondholders is presented as follows:

		A Ltd	NewCo
		Rm	Rm
Dividend income		146.0	67.2
Less: personal tax	41%	-59.9	-27.5
Dividend income after personal taxes	A	86.1	39.6
Interest income		0.0	108.0
Less: personal tax on interest	41%	0.0	-44.3
Interest income after personal taxes	B	0.0	63.7
TOTAL INCOME AFTER TAX	A+B	86.1	103.3

Page 14-25 Therefore, when personal tax rates on dividend income and interest income are equal, then it pays to use debt financing as the total income due to the shareholders and bondholders is higher with the use of debt. Let's now change tax rates. Dividends will be subject to a final dividend tax of 20%. We will assume that interest remains taxable at 41%. If this is the case, then there is a very slight disadvantage to equity financing. This is indicated as follows:

	A Ltd	NewCo

		Rm	Rm
Dividend income		146.0	67.2
Less: personal tax	20%	-29.2	-13.4
Dividend income after personal taxes#	A	116.8	#53.7
# Note rounding effect as actual numbers are 67.160-13.435 = 53.725			
Interest income		0.0	108.0
Less: personal tax on interest	41%	0.0	-44.3
Interest income after personal taxes	B	0.0	63.7
TOTAL INCOME AFTER TAX	A+B	116.8	117.4

In South Africa, pension and retirement funds will not pay tax on interest income nor will these funds pay tax on dividend income. Therefore, if bonds are issued to pension funds, then debt will retain its advantage. In this case we will increase total income from R146m to R175.2m by using debt which is equal to the income after corporate tax.

Individual investors will have varying marginal tax rates and therefore it is difficult to analyse what the net effect will be but it is safe to conclude that debt has a positive impact on the value of the firm, although by not as much as if there were no personal taxes. In the USA, it has been estimated that the net effect is that firms will be able to increase firm value by 7%, if firms increased their use of debt.

10 CAPITAL STRUCTURES IN SOUTH AFRICA AND AROUND THE WORLD

Most companies in South Africa employ a target debt–equity ratio. In a survey by Correia and Cramer (2008), 79% of companies indicated that they did apply some form of a target debtequity ratio. Yet, the survey found that close to 75% of South African companies had a target debt–equity ratio that is less than 40%.

This seems low in relation to what we would expect in terms of the trade-off theory. Companies seem to be some distance away from default with such low debt–equity ratios. High levels of profitability in the South African economy and limited prospects for expansion may explain the low levels of gearing. Management may also remember the financial crisis in 1998 when local interest rates rose to 24%. Interest rates remain high in 2024 relative to other developed economies. It is also true that the low use of debt resulted in South African companies being less affected by the financial crisis in 2008 and early 2009 relative to companies in other countries.

Page 14-26 De Wet (2006) sets out a table of capital structures for selected countries and we have depicted long-term debt to total capital ratios in Figure 14.6.

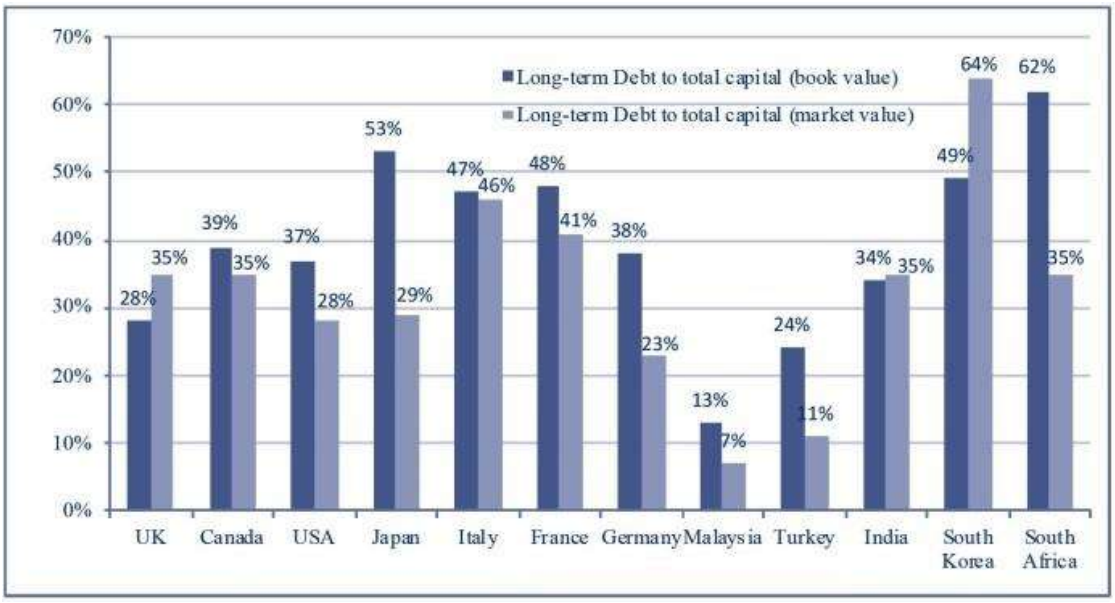


Figure 14.6 Debt to Capital ratios of firms in selected countries (Source: De Wet (2006))

Generally, firms reported relatively conservative and low debt ratios. De Wet (2006) also reported significant differences in debt to capital ratios for different industries. Companies in the Technology sector had debt to total capital (market value) ratios of only 20% whilst for Healthcare it was 33% and for Basic Materials it was 48%. The McKinsey Global Institute in 2018 reported that non-financial corporate debt mainly in the form of corporate bonds had risen from $25 trillion in 2000 to $66 trillion in 2017 and the emerging markets of China, Brazil and India were particularly at risk of future defaults[10]. In 2022, the IMF estimated that corporate debt was close to 100% of GDP. A study by De Wet and Gossel (2016) found that chief financial officers in South Africa were equally likely to follow the pecking order and trade-off theories of capital structure[11]. A study by Ramjee and Gwatidzo (2012) also suggested that South African firms follow both the pecking order and trade-off theories of capital structure and companies move toward their target capital structures relatively quickly[12].

Professor Damodaran of New York University maintains data sets, which he obtains from Value Line, Bloomberg and Capital IQ. We have used his data set of global market debt to capital ratios (adjusted for leases) to set out these ratios per industry as at January 2023. Appendix 14.1 sets out the debt to capital and debt to equity ratios for industry sectors.

As expected, we can see that the lowest use of debt is by computer software firms, biotechnology and healthcare firms while the highest use of debt (if we exclude real estate and the financial sector) is by air transport firms, power and general utilities, as well as construction, telecoms and capital-intensive sectors. Real estate, financial

services and banks have very high debt to capital ratios, which often reflect the nature of these sectors.

If we exclude the real estate and financial sectors and evaluate the debt–equity ratios rather than debt to capital ratios above (see Appendix 14.1) we will find that 62 of the 85 industry sectors reported an average market debt–equity ratio below 50%. Apart from the utilities, air transport, construction, telecoms and a few capital-intensive sectors, debt–equity ratios are reasonable. There is also a significant difference between the debt to capital ratios based on book values and the debt to capital ratios based on market values. The high gearing Page 14-27 in the financial sector compounded the consequences of the global financial crisis. Most other industry sectors had relatively low debt–equity ratios at the time of the global financial crisis, which enabled them to have the resilience to survive the economic cycle.

FROM THE REAL WORLD ...

Hudaco

Hudaco imports and distributes industrial and electronic products. Sales amounted to R8.15 billion in 2022. In relation to its capital structure and gearing, the 2022 Integrated Report stated:

> Hudaco has the capacity to take on more senior debt and our acquisition strategy may create the opportunity to do this in future, so we consider it prudent to have facilities available. Perhaps more important than managing gearing is an objective to ensure that interest on senior debt is covered at least five times by operating profit. We also aim to operate with EBITDA being at least 50% of net senior debt. Our covenants with the banks are less onerous, with the most stringent being four times interest cover and EBITDA 40% of net senior debt.
>
> In setting the ideal mix between debt and equity, the group seeks to optimise its return on shareholders' equity while maintaining prudent financial gearing. Generally, the objective is to operate with net interest-bearing debt not exceeding 2.0 times EBITDA. Excess capital will be returned to shareholders in the form of special dividends or share buy-backs when appropriate. In setting the maximum amount of unsubordinated debt the group would carry, the group's objective would also be to have net interest covered at least five times by operating profit; net interest being interest paid on interest-bearing debt less interest received. In 2022 it was 17.8 times operating profit and 20.8 times EBITDA.

Sasol has followed a policy of managing its capital structure by setting a target debt–equity range of 20–40%. Sasol also aims to maintain a net debt to EBITDA of < 1.5 times. Sasol states the following in its 2022 integrated report:

> The group and company's objectives when managing capital (which includes share capital, borrowings, working capital and cash and cash equivalents) are to maintain a flexible capital structure that reduces the cost of capital to an acceptable level of risk and to safeguard the group and company's ability to continue as a going concern while taking advantage of strategic opportunities in order to grow shareholder value sustainably. The group and company manage the capital structure and make adjustments to it in light of changes in economic conditions and the risk characteristics of the underlying assets. In order to maintain the capital structure, the group and company may adjust the amount of dividends paid to shareholders, return capital to shareholders, issue new debt, issue new debt to replace existing debt with different characteristics and/or sell assets to reduce debt. Financing risk refers to the risk that financing of the group and company's net debt requirements and refinancing of existing borrowings could become more difficult or more costly in the future. This risk can be decreased by managing the group and company within the targeted gearing ratio, maintaining an appropriate spread of maturity dates, and managing short-term borrowings within acceptable levels.

In 2023, Sasol's net debt to EBITDA was 1.2 which was well below the debt covenant threshold of 3 times. Sasol states that it is "committed to continue to reduce leverage and absolute debt levels."

Figure 14.8 depicts Sasol's actual net debt to equity ratio in relation to the target range. Page 14-28

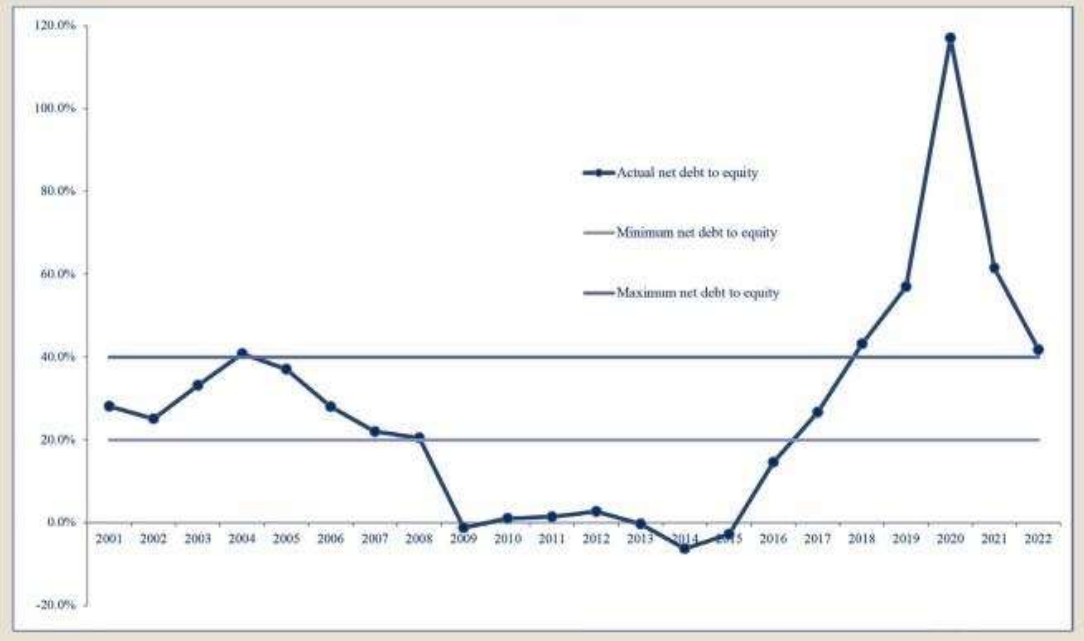


Figure 14.8 Actual and target net debt–equity range for Sasol until 2023

The increase in D/E for Sasol was due to significant capital expenditure of R179 billion over the preceding 3 years to 2018, mainly for the Lake Charles chemicals project in the USA and the PSA gas project in Mozambique. Asset sales and cost cutting enabled Sasol to significantly reduce its net debt to equity ratio. Otherwise, Sasol would have been forced to

make a rights issue to bolster its equity reserves in order to bring its capital structure closer to its target range.

In its 2022 Integrated Report, **Woolworths** indicates that its targeted leverage ratio is 1.5. Woolworths defines its leverage ratio as net debt/EBITDA.

Our analysis of South African companies indicates that companies do apply target debt–equity ratios, target debt to EBIDTA ratios and target interest coverage ratios. Capital structure management is seen as critical to manage risk and optimise a firm's return on capital.

11 EDCON: CAPITAL STRUCTURE AND VALUATION OF TAX SHIELDS[13]

Edcon was the largest clothing, footwear and textiles retailing group in South Africa with over 1 300 stores. In 2007, Bain Capital acquired all the ordinary shares in Edcon for R25 billion. This was the largest ever private equity transaction in South Africa involving a US private equity firm and was structured in the form of a leveraged buy out (LBO). Whilst we will come back to analyse this transaction in greater detail in Chapter 17, what we wish to analyse here is the impact that the leveraged buyout had on the capital structure of the group – pre-LBO and post-LBO – as well as to provide an estimate of the value of the interest tax shields created by the acquisition by Bain Capital.

Bain Capital made an offer of R46 per share. This was significantly above the JSE share price of R30.40 prior to the buy-out announcement. Edcon was delisted from the JSE after the acquisition. The buy out was structured effectively with about R19 billion of debt and R6 billion of equity (structured mainly in the form of a shareholder loan). Page 14-29

The capital structure changed significantly with the acquisition and the debt–equity ratio increased significantly after the LBO transaction.

EDCON	Pre-LBO	Post-LBO
Debt to Equity ratio	14%	394%

We have included the shareholder loan as part of equity, in determining the debt–equity ratio, post-LBO.

What were the estimated tax shields arising from the buy-out? We have done some rough estimates of the projected financing costs based on interest rates on the Euro Notes issued by Edcon which were listed on the Irish Stock Exchange, which are

hedged back into Rands at fixed forward exchange rates and other relevant interest rates. The estimated tax shields are presented in the following table:

(R millions)	2008	2009	2010	2011	2012	2013	2014	2015
Gross financing costs	3,074.0	3,140.0	2,485.1	2,726.0	2,905.5	3,093.6	3,313.4	2,821.6
Tax shield	860.7	879.2	695.8	763.3	813.5	866.2	927.7	790.0
NPVR 4,236								

The tax shields are discounted at Edcon's estimated interest rate of 11%. The number of ordinary shares in issue at the time was 490m shares and so the present value of the estimated tax shields amounts to R8.64 (4236/490) per share. We need to state however that the gross financing costs represent rough estimates at the time. It includes interest on the shareholder loan. Edcon restructured its debt post-LBO and took advantage of weak credit markets to buy back a portion of the notes listed on the Irish Stock Exchange at a substantial discount to the nominal value.

We have assumed tax shields for a period until 2015. If the loans are continually refinanced, then the company would be able to increase the value of the tax shields to about R7 billion (R25bn × 0.28).[14]

The share price of Edcon was about R30.40 prior to the announcement of the LBO. The average valuation by analysts of Edcon at the time was R37 per share. Therefore if we add our calculation of the value of the tax shields, we get to R45.64 which is very close to the offer price of R46. Of course, we would expect Edcon to be able to generate operating cost savings from the acquisition and if the company continues to refinance its loans, then this would further increase the value of the company's equity.

In 2010, the group reported an EBITDA of R2.5bn yet the group reported a loss of R926m for the year after depreciation, amortisation and financing costs. Net financing costs amounted to R2 588m in 2010.

We have determined the value of the tax shields relating to the buyout of Edcon and evaluated the offer price relative to the value of the tax shields. It seems to make sense. Unfortunately, the company has experienced losses after financing costs since the buyout.

Our analysis was undertaken at the time of the buyout. At the time, we estimated that Edcon would have seen a reduction in its WACC from about 13.7% to about 12.3%. We used a target debt–equity ratio of 316%, which resulted in the firm's beta increasing to 2.87 as compared to 0.90 prior to the buyout. The cost of debt included a spread of 2.5% (250 basis points) reflecting the BBB status of the notes issued at the

time. The financial crisis resulted in a dramatic increase in this spread but this did not directly affect Edcon as the notes were listed in a boom year for credit markets.

However, by 2014, the financial situation at Edcon had deteriorated. Edcon refinanced Page 14-30 about R4 billion of its debt, which it was only able to do so at an interest rate of JIBAR plus a margin of 7%. This is a very high cost of debt when other companies were able to borrow at about 1.7% to 2.1% above JIBAR. Secondly, operating income declined significantly and, combined with higher financing costs, resulted in material losses. Tax shields have little value if the company is incurring losses. Even here, Edcon was subject to a SARS investigation in regard to the deductibility of interest relating to the Bain acquisition of Edcon. The ensuing agreement with SARS resulted in a reported loss of future tax shields amounting to about R9 billion. That is a lot of tax shields! There were more concessions but Edcon wanted to wrap this up to enable it to undertake a listing if circumstances become favourable again. Due to its financial constraints, Edcon was required to sell its receivables book to Barclays South Africa (ABSA) for R8.8 billion. Figure 14.9 depicts operating income (EBIT), which had fallen from over R2.3 billion in 2008 to R1.3 billion in 2014. However, even more serious was the effect of its high interest charge resulting in net losses of billions of Rands each year. By 2016, the situation had deteriorated further and Edcon reported a net loss of R7 495m (before impairments). Edcon was in effect facing the real prospect of corporate failure. In the 2017 financial year, Edcon reported that the creditors would acquire a significant shareholding and control in exchange for a restructuring of Edcon's extensive debt obligations. This resulted in a reduction of interest-bearing debt from about R30bn in 2016 to about R4bn in 2017. Edcon continued to experience a challenging operating environment even though its financial position had been greatly improved due to the financial restructuring. In our view, Edcon management was so focused on managing its towering financing obligations that there may have been a lack of investment and focus on operations. In 2020, Edcon was placed under business rescue. Edcon proves the point that debt is risky and lots of debt is even worse.

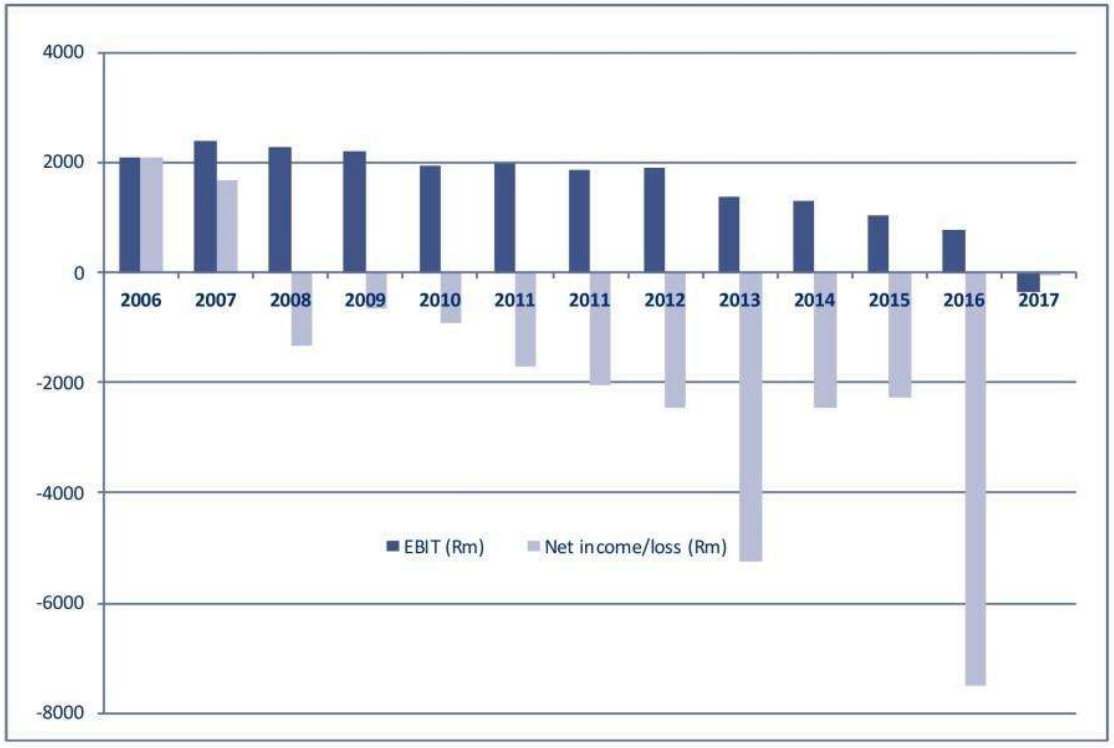


Figure 14.9 Financial leverage and losses at Edcon Page 14-31

12 CAPITAL STRUCTURE DECISIONS – A BEHAVIOURAL FINANCE PERSPECTIVE

How does management act when there are irrational investors? If management views the share price as overvalued due to market exuberance and irrational investors, then management should issue more shares. This will tilt the capital structure of the firm towards equity. When the share price is too low, then management should engage in borrowings and undertake share buy-backs, thereby tilting the firm's capital structure towards debt. The evidence is that firms do issue shares when market values are high. Firms with high valuations do tend to issue more equity and firms with low valuations tend to undertake share buy-backs. Surveys of CFOs have indicated that management will engage in market timing practices, so the amount by which shares are undervalued or overvalued is an important factor when considering the issuing of equity.

Irrational investors should not affect the investment decisions of firms. However, this assumes that firms are not dependent on new equity financing to finance growth. Firms experiencing high growth may need to finance growth with debt and equity issues. If share prices are overly depressed due to irrational investors, then this will mean that a firm will not be able to issue equity and may be forced to forego positive NPV projects. Conversely, investors may be overly confident about a firm's prospects

and may push management to take on projects that will destroy value. There is some evidence that management will use equity issues to bolster the balance sheet and increase cash balances rather than use this to fund new investments. However, other studies have indicated that overvalued firms tend to invest more than undervalued firms. There is also evidence that rights (equity) issues result in negative returns, as investors will often attribute new issues to be a signal that management considers the firm's shares to be overvalued.

We have assumed that managers are rational. What if managers are irrational? What impact will this have on capital structure decisions? Due to over-optimism, management may view the firm's share price to be undervalued. Management will avoid issuing new equity and would rather undertake borrowings or retained earnings to finance expansion. Management may follow a pecking order framework to manage the capital structure of the firm. Overoptimism may also lead management to undertake share buy-backs.

13 FINANCIAL LEVERAGE AND DURATION RISK

We have generally equated a high debt to equity ratio to a high level of risk. This is mostly true. However, we need to also consider the nature of the assets of the firm, its level of operating leverage, its business model, the nature of its cash flows and market position. A company with stable cash flows and in a strong competitive position will be able to access and utilise a higher level of debt.

Furthermore, the duration of a firm's assets should be aligned with the duration of the firm's liabilities. For example, ABC Ltd may have a debt to equity ratio of 100% and XYZ Ltd may have a debt to equity ratio of 50% and yet XYZ may be subject to a higher level of financial risk because it has long-term assets and is utilising short-term borrowings to finance these assets. XYZ Ltd may not be able to refinance its short-term borrowings when they come due and therefore may be forced to dispose of its assets in order to survive, issue new shares at a low price resulting in significant dilution or may be placed into bankruptcy or business rescue. ABC Ltd is using long-term borrowings to finance long-term assets and therefore will not be subject to a liquidity event or refinancing risks. In other cases, a collapse in a firm's operating profitability and/or a downgrade in the rating of its debt may result in the early call of its long-term debt as the firm may have contravened its loan covenants which may then make its long-term loans or bonds immediately payable. It is important to match

the duration of a firm's assets to its liabilities. We will explain duration in relation to banks in greater detail in Chapter 18. Page 14-32

SUMMARY

The choice of capital structure will depend on the financial risk the company is prepared to take, which is influenced by the other risk factors affecting the company. These are, primarily, the degree of operating and business risk, and the nature of the debt. The use of leverage will increase the return on shareholders' equity but, at the same time, it will also increase risk. There are conflicting theories regarding an optimal capital structure. On the one hand the practical approach suggests that, up to a point, the WACC decreases with increased use of debt while, at the other extreme, the pure MM approach contends that capital structure does not affect the WACC. However, it is possible to reconcile the practical approach with the MM approach when taxes, costs of financial distress, and related costs are taken into account. The pecking order theory suggests that management will raise funds from sources that retain flexibility and control. Inflation will have an impact on the optimal capital structure as the real after-tax cost of debt decreases with higher rates of inflation. Agency issues can lead to conflicts between shareholders and bondholders. This is accentuated when a firm is in financial distress or is highly levered. Interest tax shields can increase the value of debt and influence a firm's capital structure decision. The chapter also presented the capital structure decisions of South African firms and analysed the Edcon transaction which resulted in significant tax shields and a significant change in Edcon's capital structure. Whatever the capital structure chosen, it should be regarded as a target. Short-term digressions to take advantage of opportunities are all part of the financial management function.

GUIDANCE ON CAPITAL STRUCTURE

By Johnathan Dillon M.Com CA(SA)

Strategic considerations when evaluating the capital structure decision

As noted in Chapter 14, capital structure theories are not intended to be rigidly applied and reasons for deviating from what may be considered an optimal capital structure were mentioned. These included financial flexibility as well as market timing theory. However, there are a number of other strategic reasons why a company may deviate from its theoretically optimal method of financing. These are discussed below and should be considered when evaluating a company's capital structure:

- ***Business risk:*** If a company, by virtue of its operations, is subject to a high level of business risk, then it will be less likely to use financial leverage. Lenders will also be less inclined to grant credit to companies with high business risk. Therefore in such an instance debt finance will be limited; however, where business risk is assessed as low then a company will be more likely to use debt finance.
- ***Control:*** If certain shareholders barely have control (e.g. 50.1%) and are not in a position to buy any more shares, then excessive use of debt finance is likely. This is often particularly relevant for family controlled companies or holding companies who are not able to follow their rights in any rights issue due to cash flow constraints.
- ***Growth rate:*** Faster growing companies generally rely more heavily on external capital. Slower growth can be financed by retained earnings; however, rapid growth requires external funds and often debt is used due to the advantages thereof e.g. debt is easier and less costly to raise, does not have control implications and is cheaper than equity finance. However, if high growth is matched with high business risk then this will lead to the increased use of equity finance.
- ***Long-run viability:*** Large companies providing essential services, such as telecommunication and electricity providers like Telkom and Eskom, have a responsibility to provide continuous service. These companies should refrain from using excessive debt Page 14-33 finance in order to ensure their long-run viability. If the cash flows are highly stable, then one may argue that this would support a higher debt ratio. However, these companies are increasingly under pressure due to changing business models. Telkom's market has been negatively and significantly affected by firms such as MTN and Vodacom. Eskom has been increasingly affected by high operating costs, corporate governance issues, capital expenditure and cost blow-outs of investing in new power stations, unpaid accounts, the role of independent electricity suppliers, lower demand and limited tariff adjustments.
- ***Management constraints (agency problem):*** If a company generates excess cash then management may follow their own interests and squander the company's money. This can be avoided by limiting excess cash through paying dividends or repurchasing shares (decreasing equity). Another alternative is to shift the capital structure to more debt as the interest payment limits the amount of available cash. Furthermore, managers faced with higher levels of financial risk and possible job losses, should the company fail, will be less likely to squander cash.
- ***Managerial conservatism:*** Well diversified investors have eliminated most, if not all, the diversifiable risk from their portfolios. Therefore, the average investor can tolerate a

relatively high probability of financial distress as the risk thereof will be offset by other investments. However, managers are more concerned about financial distress and will be inclined to use less debt than the average investor desires as they aim to protect their jobs.

- ***Market conditions (market timing theory):*** Conditions in equity and debt markets undergo both long-run and short-run changes that can affect a company's optimal capital structure. When equity markets are rallying or a company considers its shares overvalued then more equity capital will be raised. During a credit crisis, as experienced in the period following the global financial crisis of 2008, low-rated companies in need of capital were forced to issue shares, regardless of their target capital structures. There may also be times when market interest rates are favourable making debt finance more attractive.
- ***Nature of assets:*** Companies with assets suitable to act as security for loans, such as property companies, tend to use more debt as the security enables debt finance to be raised more easily and on favourable terms. On the contrary, companies with minimal tangible assets, such as pharmaceutical companies engaged in the research and development of new drugs, utilise relatively little debt.
- ***Profitability:*** Companies with high rates of return on investment (and with limited growth opportunities) tend to use less debt funding. High profits enable a company to finance its operations with retained earnings.
- ***Rating agency and lender attitudes:*** Rating agencies and lenders' attitudes are important determinants of financial structure. Often management discuss the capital structure of their company with these parties. This is done in order to ensure that the company's capital structure is acceptable to lenders and rating agencies which should secure a good credit rating, cheaper debt and available lines of credit.
- ***Reserve borrowing capacity (financing flexibility):*** Maintaining a reserve borrowing capacity preserves a company's ability to issue debt on favourable terms and when needed e.g. when interest rates are low or a low risk investment opportunity is available. In order to maintain this borrowing capacity, companies generally use less debt than they would otherwise do.
- ***Stakeholder theory:*** Stakeholders other than those who have provided finance, such as customers, suppliers and employees, are less willing to transact with or work for a company that has high levels of debt. When a company has "too much" debt, stakeholders may rather commit to doing business with or working for another company.
- ***Taxation:*** Interest is a deductible expense but dividends are not deductible so the higher the tax rate, the greater the advantage of using debt. However, if a company is in a tax

loss position then the advantage of using debt is deferred. The tax situation of a company therefore influences its level of debt financing. Page 14-34

CAPITAL STRUCTURE ANALYSIS

Financial statement ratio analysis was covered in Chapter 5. Debt management ratios are an important category of ratios in the context of this chapter (capital structure). The following debt management or gearing (leverage) ratios are useful to calculate and consider when analysing the capital structure or financial risk of an entity:

Note: *There could be numerous variations of these debt related ratios e.g. the debt ratio could be "long-term debt" divided by "net assets" or debt could be defined as "net debt" which is debt less any cash reserves. Debt often will only refer to interest-bearing debt.*

$$\text{Debt ratio} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

$$\text{Debt to equity} = \frac{\text{Total Debt}}{\text{Total Equity}}$$

$$\text{Cash flow to total debt} = \frac{\text{Cash flow from operations}}{\text{Total Debt}}$$

$$\text{Times interest earned (interest cover)} = \frac{\text{Earnings before interest and tax (EBIT)}}{\text{Interest expense}}$$

$$\text{EBITDA interest coverage} = \frac{\text{EBIT + Depreciation + Amortisation (EBITDA)}}{\text{Interest expense}}$$

$$\text{Fixed charge coverage} = \frac{\text{EBITDA + Lease Payments}}{\text{Interest expense + Lease Payments}}$$

Note: *Interest in the above ratios usually comprises only interest expense with interest income added to EBIT; however, interest income can be offset against interest expense – this is then termed net interest expense.*

$$\text{Degree of Financial Leverage (DFL)} = \frac{\text{EBIT}}{\text{EBIT – Interest expense (NPBT)}}$$

Note: *DFL is an indicator of financial risk – the greater the number, the higher the interest charges and the greater the financial risk (refer to Chapter 3).* Page 14-35

APPENDIX 14.1 Global Debt and Debt–equity ratios

Note: Capital refers to debt plus equity and these will be either stated at book values or market values

Industry Name	*No. of firms*	*Book Debt to Capital*	*Market Debt to Capital (Unadjusted)*	*Market Debt to Capital (adjusted for leases)*	*Market D/E (unadjusted)*	*Market D/E (adjusted for leases)*	*Interest Coverage Ratio*	*Debt t EBITD*
Advertising	362	45.54%	26.36%	26.56%	35.80%	36.16%	5.52	3.66
Aerospace/Defense	278	49.89%	18.60%	18.86%	22.85%	23.24%	4.64	3.59

Industry Name	*No. of firms*	*Book Debt to Capital*	*Market Debt to Capital (Unadjusted)*	*Market Debt to Capital (adjusted for leases)*	*Market D/E (unadjusted)*	*Market D/E (adjusted for leases)*	*Interest Coverage Ratio*	*Debt t EBITD*
Air Transport	155	76.27%	52.24%	52.38%	109.38%	109.98%	0.31	223.8(
Apparel	1160	39.52%	15.25%	15.36%	18.00%	18.15%	16.49	2.19
Auto & Truck	154	48.77%	42.41%	42.43%	73.65%	73.69%	13.28	4.94
Auto Parts	746	32.94%	26.09%	26.06%	35.29%	35.24%	5.19	2.67
Beverage (Alcoholic)	220	36.83%	14.02%	14.07%	16.31%	16.38%	9.46	2.42
Beverage (Soft)	100	49.80%	13.05%	13.33%	15.01%	15.38%	9.49	2.75
Broadcasting	135	39.22%	44.27%	44.64%	79.45%	80.64%	4.51	3.25
Building Materials	454	37.74%	20.12%	20.18%	25.19%	25.28%	12.26	2.19
Business & Consumer Services	961	44.97%	18.68%	18.87%	22.98%	23.26%	7.36	3.03
Cable TV	54	64.78%	48.80%	49.45%	95.32%	97.83%	4.07	3.88
Chemical (Basic)	879	36.64%	27.00%	27.05%	36.98%	37.08%	9.22	2.21
Chemical (Diversified)	68	36.96%	34.27%	34.29%	52.14%	52.18%	11.17	2.24
Chemical (Specialty)	922	34.46%	17.38%	17.46%	21.04%	21.15%	12.52	2.13
Coal & Related Energy	212	25.60%	19.55%	19.56%	24.30%	24.31%	19.44	1.10
Computer Services	1105	36.63%	13.91%	14.05%	16.16%	16.35%	13.95	2.00
Computers/Peripherals	333	38.17%	10.61%	10.59%	11.87%	11.85%	22.35	1.20
Construction Supplies	790	38.97%	31.06%	31.13%	45.06%	45.21%	5.34	3.35
Diversified	314	40.30%	35.74%	35.86%	55.62%	55.90%	2.79	3.31
Drugs (Biotechnology)	1267	39.27%	11.30%	11.40%	12.74%	12.86%	2.74	4.31
Drugs (Pharmaceutical)	1352	36.02%	12.89%	13.15%	14.80%	15.14%	11.86	2.03
Education	251	37.55%	24.48%	24.74%	32.41%	32.88%	2.19	4.55
Electrical Equipment	1045	32.10%	14.30%	14.30%	16.69%	16.69%	7.23	3.15
Electronics (Consumer & Office)	134	36.53%	30.72%	30.72%	44.33%	44.33%	9.17	2.36
Electronics (General)	1457	29.28%	16.31%	16.34%	19.49%	19.53%	10.80	1.96
Engineering/Construction	1269	49.05%	51.02%	51.04%	104.16%	104.24%	3.92	5.61
Entertainment	752	37.63%	21.20%	20.88%	26.91%	26.40%	3.36	4.53
Environmental & Waste Services	370	51.48%	27.17%	27.23%	37.32%	37.41%	3.89	4.39
Farming/Agriculture	426	49.97%	31.71%	31.71%	46.44%	46.44%	8.27	4.45
Food Processing	1397	41.93%	22.41%	22.48%	28.89%	28.99%	6.53	3.12
Food Wholesalers	169	59.19%	40.16%	40.47%	67.10%	67.97%	3.21	5.36
Furn/Home Furnishings	362	36.03%	20.37%	20.43%	25.57%	25.68%	9.43	2.25
Green & Renewable Energy	248	54.48%	38.36%	38.39%	62.22%	62.31%	2.55	5.77
Healthcare Products	896	31.93%	11.52%	11.62%	13.01%	13.15%	10.52	2.31

Industry Name	*No. of firms*	*Book Debt to Capital*	*Market Debt to Capital (Unadjusted)*	*Market Debt to Capital (adjusted for leases)*	*Market D/E (unadjusted)*	*Market D/E (adjusted for leases)*	*Interest Coverage Ratio*	*Debt t EBITD*
Healthcare Support Services	460	43.13%	21.15%	21.48%	26.83%	27.36%	7.59	3.11
Heathcare Information & Tech.	447	33.94%	10.54%	10.66%	11.79%	11.93%	10.43	2.70
Homebuilding	171	30.28%	28.05%	31.53%	38.99%	46.06%	44.83	1.99
Hospitals/Healthcare Facilities	231	65.27%	33.40%	33.98%	50.14%	51.48%	3.28	4.85
Hotel/Gaming	650	63.73%	35.91%	36.06%	56.04%	56.39%	0.85	16.74
Household Products	589	42.05%	12.16%	12.29%	13.84%	14.01%	13.90	2.18
Information Services	242	42.13%	12.16%	12.31%	13.84%	14.04%	9.24	2.78
Insurance (General)	206	41.78%	29.06%	29.19%	40.96%	41.22%	7.72	3.02
Insurance (Life)	142	55.27%	49.55%	49.62%	98.23%	98.48%	4.30	6.57
Insurance (Prop/Cas.)	235	28.22%	19.79%	20.18%	24.67%	25.28%	7.54	1.87
Machinery	1463	31.94%	14.59%	14.72%	17.09%	17.26%	10.27	2.24
Metals & Mining	1783	32.62%	20.96%	21.01%	26.52%	26.60%	14.80	1.33
Office Equipment & Services	144	37.60%	25.24%	25.13%	33.76%	33.57%	6.11	2.93
Oil/Gas (Integrated)	36	27.76%	16.08%	16.27%	19.17%	19.43%	28.34	1.11
Oil/Gas (Production & Exploration)	616	31.11%	20.70%	20.77%	26.10%	26.21%	17.04	1.41 Pa 14-36
Oil/Gas Distribution	166	57.03%	42.22%	42.27%	73.08%	73.21%	2.66	6.08
Oilfield Svcs/Equip.	455	41.50%	31.60%	31.78%	46.20%	46.59%	7.80	3.69
Packaging & Container	414	48.64%	32.80%	32.86%	48.80%	48.94%	6.15	3.26
Paper/Forest Products	268	39.68%	37.99%	38.04%	61.27%	61.39%	8.25	2.56
Power	485	58.31%	48.97%	49.02%	95.95%	96.15%	0.30	5.99
Precious Metals	930	22.47%	15.39%	15.43%	18.19%	18.25%	9.96	1.22
Publishing & Newspapers	327	27.72%	22.49%	22.99%	29.01%	29.86%	6.48	2.78
Recreation	323	45.26%	24.32%	24.70%	32.13%	32.81%	4.71	3.93
Reinsurance	34	32.58%	25.75%	26.02%	34.69%	35.16%	1.68	3.81
Restaurant/Dining	382	82.13%	24.21%	24.61%	31.95%	32.65%	5.06	5.12
Retail (Automotive)	193	64.49%	36.71%	36.89%	58.00%	58.44%	6.11	4.26
Retail (Building Supply)	98	80.12%	19.21%	19.39%	23.78%	24.05%	12.40	2.34
Retail (Distributors)	1006	51.09%	40.87%	40.91%	69.13%	69.22%	6.53	5.12
Retail (General)	189	54.33%	26.20%	26.73%	35.50%	36.48%	5.83	3.32
Retail (Grocery and Food)	181	56.39%	35.94%	36.67%	56.11%	57.90%	4.98	3.76
Retail (Online)	342	47.49%	17.57%	17.04%	21.32%	20.54%	0.20	5.15

Industry Name	*No. of firms*	*Book Debt to Capital*	*Market Debt to Capital (Unadjusted)*	*Market Debt to Capital (adjusted for leases)*	*Market D/E (unadjusted)*	*Market D/E (adjusted for leases)*	*Interest Coverage Ratio*	*Debt t EBITD*
Retail (Special Lines)	495	51.09%	23.68%	23.74%	31.03%	31.13%	7.73	3.33
Rubber & Tires	89	36.28%	33.86%	33.86%	51.19%	51.19%	6.13	2.39
Semiconductor	624	27.41%	10.37%	10.40%	11.58%	11.61%	20.44	1.20
Semiconductor Equip	342	27.47%	6.59%	6.62%	7.06%	7.09%	28.87	1.09
Shipbuilding & Marine	349	30.16%	32.35%	33.34%	47.83%	50.02%	21.59	1.53
Shoe	85	39.47%	11.04%	11.03%	12.41%	12.39%	13.75	2.39
Software (Entertainment)	320	20.60%	6.89%	6.73%	7.39%	7.21%	38.19	0.86
Software (Internet)	152	43.41%	14.27%	14.29%	16.65%	16.67%	-0.83	10.99
Software (System & Application)	1648	37.57%	8.17%	8.21%	8.90%	8.94%	10.50	2.33
Steel	710	32.79%	32.07%	32.12%	47.22%	47.33%	12.32	1.33
Telecom (Wireless)	99	49.61%	41.26%	41.00%	70.25%	69.48%	4.01	2.91
Telecom. Equipment	461	31.45%	12.27%	12.27%	13.99%	13.99%	9.11	2.06
Telecom. Services	295	52.81%	45.50%	45.52%	83.48%	83.56%	4.44	3.04
Tobacco	56	48.29%	21.08%	21.15%	26.71%	26.82%	11.30	2.14
Transportation	302	46.55%	31.26%	31.15%	45.47%	45.23%	7.17	3.23
Transportation (Railroads)	50	49.28%	27.80%	28.09%	38.51%	39.07%	6.41	4.59
Trucking	220	57.15%	36.61%	37.03%	57.76%	58.80%	4.07	5.45
Utility (General)	51	56.40%	44.59%	44.65%	80.46%	80.66%	3.76	5.34
Utility (Water)	104	55.35%	45.24%	45.26%	82.63%	82.68%	2.72	6.08
Total Market (without financials)	42593	**43.64%**	**25.38%**	**25.51%**	**34.01%**	**34.24%**	**7.13**	**3.17**

[Source: A. Damodaran NYU, Bloomberg and Capital IQ, January 2023]

SELF-STUDY PROBLEM

Page 14-37 Consider two firms, identical except for their capital structure. Both will earn R300m in a boom and R100m in a recession. There is a 50% chance of each event. Firm U is financed entirely by equity. Its shares are valued at R1 000m. Firm L has issued R800m of risk-free debt at an interest rate of 10%, and therefore R80m of income is paid out as interest. There are no taxes or other market imperfections. Investors can borrow and lend at the risk-free rate of interest.

Applying MM:

(i) What is the value of L's shares?

(ii) Suppose that you invest R40m in U's shares. Is there an alternative investment in L that would give identical payoffs in a boom and a recession? What is the expected payoff from such a strategy?

(iii) Now suppose that you invest R40m in L's shares. Design an alternative strategy with identical payoffs.

(iv) What does this say about capital structure?

Solution to self-study problem

(i) Because the firms are identical except for financial structure, and there are no taxes or other market imperfections, the total value of these companies must be the same. Thus L's shares are worth R1 000m – R800m = R200m.

(ii) If you own R40m of U's ordinary shares, you own 4% of the issued shares and thus are entitled to 0.04 × R300m = R12m if there is a boom and 0.04 × R100m = R4m, if there is a recession. The equivalent investment is to purchase 4% of L's issued shares, which will cost 0.04 × R200m = R8m, and to invest R32m in the risk-free rate. The total invested is the same (R40m); in a boom you are entitled to 0.10 × R32m + 0.04 × (R300m – R80m) = R12m, and in a recession you are entitled to 0.10 × R32m + 0.04 × (R100m – R80m) = R4m.

(iii) If you own R40m of L's shares, you own 20% of the issued shares and are thus entitled to 0.20 × (R300m – R80m) = R44m if there is a boom and 0.20 × (R100m – R80m) = R4m, if there is a recession. The equivalent investment is to purchase 20% of U's issued shares, which will cost 0.20 × R1 000m = R200m, and to borrow R160m at the risk-free rate. The total invested is the same at R40m; in a boom you are entitled to –0.10 × R160m + R0.20 × R300m = R44m, and in a recession you are entitled to –0.10 × R160m + 0.20 × R100m = R4m.

(iv) Given the restrictive assumptions, this shows that capital structure is irrelevant. In all the situations, earnings were the same irrespective of the capital structure. In practice the assumptions are not likely to apply. Page 14-38.

QUESTIONS

Question 14.1

Why might companies in the same industry have different capital structures?

Question 14.2

Debt is generally believed to be a less expensive form of finance than equity. Why, then, do companies not use more debt in their capital structure?

Question 14.3

How does inflation impact on the real after-tax cost of debt? Discuss any assumptions made in arriving at your analysis. Does inflation have an impact on the level of financial risk? Construct a simple example to support your analysis.

Question 14.4

Set out the pecking order a company should follow in raising money.

Question 14.5

A company has invested in a project that cost R20m and returns a cash flow of R5 056 710 before tax each year for the 10 years of its life. As this return exactly matches the company's required rate of return they are prepared to fund it. The cost of debt is 12% and the tax rate 27%. If the debt to equity ratio is 1:2 what is the return on equity (ROE)?

Question 14.6

A company wants to restructure its balance sheet to increase the debt to equity ratio. Suggest 3 ways that this can be done.

Question 14.7

Discuss (with the aid of diagrams) the following approaches to capital structure theory:

(i) net operating income;

(ii) MM, first with company tax alone, then with agency and other related costs.

Question 14.8

The government is considering lowering the company tax rate from 27% to 20%. What effect will this have on the debt ratios of firms? What if the government is also considering removing the tax deductibility of interest?

Question 14.9

Prepare a briefing document for a board of directors discussing issues that might influence a company's capital structure strategy.

(ACCA)

Question 14.10

Thelma Xoli is about to start an earth-levelling operation for which she requires capital of about R100m to buy the equipment and operate the company. Earnings before tax and interest are expected to amount to R20m (tax rate is 27%). The funds can be raised by issuing shares for the full amount or, alternatively, half the requirements may be raised through the issue of 12% debentures.

Required:

Calculate the return on equity under both alternatives (ignore issue expenses). Which factors should be considered in deciding on which financing alternative to adopt?

Page 14-39

Question 14.11

Company A and Company B are in the same risk class, and are identical in every respect except that Company A is levered while Company B is not. Company A has R300m in 5% debentures. Both firms earn 10% before interest and taxes on their R500m of total assets. Assume perfect capital markets, rational investors, a company tax rate of 27%, and a capitalisation rate of 10% for an all-equity company.

Required:

(a) Compute the value of each firm using the net income (NI) approach.

(b) Compute the value of each firm using the net operating income (NOI) approach.

(c) Using the NOI approach, calculate the after-tax weighted-average cost of capital, k, for firms A and B. Which of these two firms has an optimal capital structure according to the NOI approach? Why?

(d) Why is the cost of equity capital higher for the levered firm, but its weighted average cost of capital lower?

(e) According to the NOI approach, the values for firms A and B computed in part (a) above are not in equilibrium. If a situation like this exists, an investor in the overvalued firm can, through the arbitrage process, secure the same income at lower cost. Assuming that you own 1% of A's shares, show the process which will give you the same amount of income but at less cost.

Question 14.12

Overseas Enterprises Inc, manufacturers of parts used in the television industry, is considering registering a company in South Africa called CTV Ltd. An in-house analyst has been assigned the task of determining the optimum capital structure of CTV. Because of the importance of the decision, management has also hired the

investment banking firm of Stanley Ltd to conduct a parallel analysis of the situation. Mr Smith, the in-house analyst, has decided to carry out the analysis using the MM framework. Ms Gamede, the Stanley Ltd consultant, who is well versed in modern financial theory and is confident of her ability to predict the firm's debt and equity costs at various levels of debt, has decided to estimate the optimal capital structure as that structure which minimises the firm's weighted average cost of capital. The following data are relevant to both analyses:

- EBIT = R40m per year, in perpetuity
- Tax rate = 27%
- Dividend ratio payout = 100%
- Current required rate of return on equity = 12%

The cost of capital schedule predicted by Ms Gamede is as follows:

At debt level of (in millions):	**R0**	**R20**	**R60**	**R100**	**R140**
Cost of debt (%)	8.0	8.0	9.0	11.0	16.0
Cost of equity (%)	12.00	12.25	13.00	13.40	17.0

The following are additional data acquired by Mr Smith:

Present value of liquidation costs = R80m					
At debt level of (in millions):	**R0**	**R20**	**R60**	**R100**	**R140**
Probability of liquidation	0	0	0.07	0.17	0.90

Required:

(a) Calculate the different values of the firm at the various levels of debt for both Mr Smith's and Ms Gamede's methods.

(b) Comment on the similarities and differences in their recommendations. Page 14-40

Question 14.13

GB Ltd has been successfully producing and marketing fruit juice from the vineyards and orchards around the Boland. Expansion and maintenance of these operations is estimated to require capital investment amounting to R63m. Such investment typically yields a return of 15% per year. A new natural spring has been discovered in the area and preliminary tests indicate the mineral waters to be of a quality to match even the best waters from France. An amount of R138m would be required to bring the spring to full production and to bottle and market the product, both locally and overseas. This

investment is expected to generate a return of 20% per year. The following details may be relevant:

Dividend per share (cents)	130	Debt ratio	40%
Number of Shares issued (millions)	35	Cost of debt	12%
Current earnings (Rmillion)	125.5	Cost of equity	15%
Market price per ordinary share	26.00	Effective tax rate	27%
Cost of issuing new shares (cents per share)	100		

Required:

(a) Assuming all the projects are undertaken, what is the cost of capital before and after the investments?

(b) Should all the projects be undertaken?

Question 14.14

Assume the same details as in 14.10, except that earnings before interest and tax are expected to be earned according to the following probabilities:

Earnings (Rm)	0	10	20	30	40
Probability	20%	20%	20%	20%	20%

Required:

Make a recommendation regarding the capital structure.

Question 14.15

It has been suggested that, if the equity and long-term capital of a company is valued at market price, and if the rate of return on that capital is calculated from earnings before loan interest and taxation is ignored, then that rate of return will be identical for all companies having the same total risk. The relevant data for two such companies are given below:

	Company A	Company B
Current value of capital:		
Number of ordinary shares	90m	150m
Market price per share	R1.20	R1.00
Market value of equity	R108m	R150m
6% secured non-redeemable debt at par	R60m	
Earnings before interest	R18m	R18m

All income after loan interest is distributed as a dividend.

Required:

(a) Explain and illustrate the process by which, through investors' actions, the market values of the two companies might be brought into equilibrium.

(b) List the further assumptions implicit in your calculations.

(c) Comment on the likely effect of taxation on the market values of the two companies. No calculations are required. Page 14-41

Question 14.16

	Co. A Rm	Co. B Rm	Co. C Rm	Co. D Rm
Equity	150	100	100	100
Debt	0	20	50	100
ROE	20%	15%	12%	15%
Kd	8%	10%	12%	15%
Tax rate	27%	27%	27%	27%

All four companies have the opportunity to invest in a project that costs R50m and returns a perpetuity of R15m per year after interest but before tax.

Which companies should take up the project and why?

Question 14.17

A profitable company has a capital structure consisting of 40% debt and 60% equity and a WACC of 12%. Tax is 27% and the interest rate is 10%.

A new project is being considered and the R40 million cost will either be raised by an issue of shares that will increase the total equity to R100 million or by an issue of debt. The new project will return a perpetuity of R4.725m pa after tax.

Required:

(a) Should the project be accepted under the present capital structure?

(b) You have been asked to assess the project and must decide whether it should be financed with either an issue of debt or equity. Assuming that the return on equity and debt do not change and the project will be undertaken which of the two financing options is best?

Question 14.18

The finance director of Kulpar Ltd is concerned about the impact of capital structure on the company's value, and wishes to investigate the effect of different capital structures. He is aware that as gearing increases the required return on equity will also increase, and the company's interest cover is likely to decrease. A decrease in interest cover could lead to a change in the company's credit rating by the rating agencies. He has been informed that the following changes are likely:

Interest cover	Credit rating	Interest rate
More than 6.5	AA	8.0%

4.0 – 6.5	A	9.0%
1.5 – 4.0	BB	11.0%

Summarised financial data:-	
	Rmillion
Net operating income	110.0
Depreciation	-20.0
Earnings before interest and tax	90.0
Interest	-22.0
Taxable income	68.0
Tax (30%)	-20.4
Net income	47.6
Capital spending (Rm)	20.0

Page 14-42 Market value of equity is R458 million, and of debt R305 million. Kulpar's equity beta is 1.4. The beta of debt may be assumed to be zero. The risk-free rate is 5.5% and the market return 14%. The company's growth rate of cash flow may be assumed to be constant, and to be unaffected by any change in capital structure. The tax rate is 30%.

Required:

(a) Determine the likely effect on the company's cost of capital and corporate value if the company's capital structure was:

(i) 80% equity, 20% debt by market values; and

(ii) 40% equity, 60% debt by market values.

Recommend which capital structure should be selected.

Note

Any change in capital structure would be achieved by borrowing to repurchase existing equity, or by issuing additional equity to redeem existing debt, as appropriate. The current total firm value (market value of equity plus market value of debt) is consistent with the growth model (CF1/(k – g)) applied on a corporate basis. CF1 is next year's free cash flow; k is the weighted-average cost of capital (WACC), and g the expected growth rate. Company free cash flow may be estimated using EBIT(1 – t) + depreciation – capital spending. State clearly any other assumptions that you make.

(b) Discuss possible reasons for errors in the estimates of corporate value in part (a) above.

(*ACCA – adapted*)

Question 14.19

Brilliant Breeze Ltd ('Breeze') is a leader in the renewable energy industry in South Africa and has a financial year end of 30 September. Breeze is listed on the AltX. The company manufactures wind turbines and constructs and maintains wind farms for its industrial customers in South Africa. The company has been expanding rapidly in recent years, using mainly debt financing, and is currently highly geared. Breeze's target debt ratio is 60%. R500m of Breeze's long-term debt is maturing in early October 20x2. Assume the current date is 30 September 20x2. Breeze does not expect to have sufficient cash available to repay this debt. The following sources of finance are being considered:

- A R500 million rights issue. The following market data is available:

Selected market data	
Yield on 364-day Treasury Bills	4.85%
Yield on Government Bonds (matures 21/12/20×6)	7.50%
Yield on Government Bond (matures 31/3/20×12)	9.75%
Market risk premium: South Africa	7.00%
Unlevered beta of Breeze	0.85
Current market price of Breeze shares	20.00

- A bank loan of R500m at a market-related fixed interest rate of 12% per annum. Interest is payable annually on 30 September. The capital portion of the loan will be repayable in five equal annual instalments commencing on 30 September 20×3. The bank will require a first cession over Breeze's accounts receivable and inventory balances to cover the capital amount outstanding.
- A green corporate bond that will have a nominal value of R550m and an issue price of R500m. The coupon will be payable quarterly in arrears at the South African Reserve Bank repurchase rate (repo rate) of 3.75% plus 4.25%, equalling 8.00%. The first coupon will be payable on 31 December 20×2. The bond will be redeemable at par on 30 September 20×9.

Page 14-43 Irrespective of the source, the finance will be raised on 1 October 20x2. The following extract from Breeze's budget is available to assist the management team with evaluating which source of finance Breeze should select:

Extracts from the budgeted statements of financial position at 30 September	Notes	20×2	20×3	20×4
		Rm	Rm	Rm
Share capital (100 million ordinary shares)	1	500	500	500
Retained earnings		1250	1325	1500
Shareholders' interest		1750	1825	2000

Extracts from the budgeted statements of financial position at 30 September	Notes	20×2	20×3	20×4
Long-term debt	1, 2	3250	3250	3250
		5000	**5075**	**5250**
Property, plant and equipment		3500	3575	3475
Current assets		2000	2125	2155
Current liabilities		(500)	(625)	(380)
		5000	**5075**	**5250**

Notes to the budget

1. The budgeted amounts include the effects of the planned expansion into the rest of Africa. However, it does not include the raising of any new finance referred to above.
2. R750 million of the long-term debt is from NatBank and matures in September 20x9. The high debt ratio resulted in a breach of one of the loan covenants contained in the NatBank loan agreement. However, Breeze was able to successfully renegotiate the loan terms. The revised loan terms stipulate that the loan will become due and payable on 1 October 20x4 if Breeze's debt ratio is not below or equal to the 60% target by 30 September 20x4.

Hamada formula: Beta (levered) = Beta (unlevered) × [1 + {(1 – T) × (debt / equity)}]

Required:

Critically evaluate the sources of finance being considered. Your evaluation should include a comparison of the costs of finance and the impact of each source of finance on the target debt ratio for all budgeted periods (round all final answers to two decimals), and any other qualitative factors not dealt with.

(*SAICA ITC 2022 – adapted*) **Page 14-44**

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[1] Equity will maintain an option value depending on the amount of time remaining until the plug is pulled and the firm is put into bankruptcy. The business rescue provisions in Chapter 6 of the Companies Act, 2008 will increase the value of this option.

[2] Edcon was later placed under business rescue.

[3] Nampak undertook a consolidation of every 250 shares into 1 share and then offered 5.714m shares at R175 to its shareholders. In order to restructure its capital structure, the company has been undertaking asset disposals, extending the term of its loans (at a higher interest rate) and undertaking a rights issue to raise equity finance.

[4] We should evaluate the after-tax required return of lenders. What is quoted by the market in terms of bond yields reflects the view of many investors with varying tax rates. However, the rate on preference share funding offered by banks was often set at about 70-75% of prime which

indicates that the banks (lenders) will reduce the required return substantially for tax-free income, being in this case preference dividends.

[5] Highly levered capital structures with significant tax shields are becoming increasingly subject to investigation by SARS, which may disallow the interest deductions in terms of tax avoidance legislation. Any possible investigation and retrospective decisions on the deductibility of interest will increase the risk of using mostly debt to finance any acquisition.

[6] After 31 March 2023, an interest deduction is limited to 30% of adjusted taxable income (tax EBITDA) where there is a controlling relationship and the interest income to the recipient is not subject to tax. We will assume in this example that the recipient is subject to tax (which will apply in most cases).

[7] You may wish to ask what A Ltd has done with R900m, for if it has simply invested in interest securities, and interest is taxable at 27%, are we not simply playing musical chairs? A Ltd may have enabled BEE participation or A Ltd may decide to pay a dividend and enable its shareholders to obtain diversification (limited to amount not provided as security). The end effect is that debt tax shields may not be as valuable as they seem, but more about that later.

[8] The bank will probably increase the interest rate due to the higher risk of non-payment and impose loan covenants which may constrain the operating flexibility of the firm. If the interest rate is raised to 11%, then the WACC will remain at 11.4% with the higher D/E ratio.

[9] De Wet, JHvH (2006). Determining the optimal capital structure: a practical contemporary approach, *Meditari*, 14(2): 1–16.

[10] McKinsey & Co, Rising Corporate Debt: Peril or Promise, *McKinsey Global Institute Discussion Paper*, June 2018.

[11] De Wet, LH and Gossel, S (2016). South African Capital Structure Decisions: A Survey of Listed Companies, *Journal of African Business*, 17(2): 1–21.

[12] Ramjee, A and Gwatidzo, T (2012). Dynamics in capital structure determinants in South Africa, *Meditari Accountancy Research*, 20(1): 52–67.

[13] We are grateful to Walter Rauch for sourcing most of the information in this section.

[14] The corporate tax rate was reduced to 27% in 2023 but this was unknown at the time of the takeover.